Organic Chemistry: Some Basic Principles and Techniques

Organic compounds keep life going on earth, and they turn up in clothing, fuels, polymers, dyes and medicines.

By the end of this unit

- say why carbon is tetravalent, and what shapes organic molecules take from it
- write the structure of an organic molecule in more ways than one
- classify the organic compounds
- name a compound by the IUPAC system, and work back from a name to its structure
- follow the idea of an organic reaction mechanism
- explain how electronic displacements bear on the structure and the reactivity of organic compounds
- tell one type of organic reaction from another
- learn the techniques for purifying organic compounds
- write the chemical reactions used in qualitative analysis, and follow the principles behind quantitative analysis

Where the unit starts from

- the previous unit showed **catenation**, the property peculiar to carbon of **bonding covalently to further carbon atoms**
- it also bonds covalently to **hydrogen, oxygen, nitrogen, sulphur, phosphorus and the halogens**
- the compounds that result are studied under a branch of their own, **organic chemistry**
- this unit gathers the **basic principles and the techniques of analysis** needed to understand how organic compounds form and how they behave

A.1 General introduction

- life on earth **could not be sustained without organic compounds**; among them stand molecules as complex as **deoxyribonucleic acid (DNA)**, which carries the genetic information, and the **proteins** our blood, muscle and skin are built of
- they appear in **clothing, fuels, polymers, dyes and medicines**, some of the important areas these compounds are applied in
- organic chemistry as a science is some **two hundred years old**
- from about **1780** chemists drew a line between the compounds **taken from plant and animal** and those **made up from mineral**
- the Swedish chemist **Berzilius** held that a **'vital force' lay behind the formation of organic compounds**
- that notion was **rejected in 1828, when F. Wohler synthesised urea**, an organic compound, from the inorganic **ammonium cyanate**

$$\underset{\text{Ammonium cyanate}}{NH_4CNO} \xrightarrow{\text{Heat}} \underset{\text{Urea}}{NH_2CONH_2}$$

- **Kolbe's acetic acid (1845)** and **Berthelot's methane (1856)**, pioneering syntheses both, settled the matter: an **inorganic source will yield an organic compound in the laboratory**

The unit runs in three parts: **A · Structure, classification and naming**, then **B · Reaction mechanism**, then **C · Purification and analysis**.

- organic chemistry took its **modern shape** once the **electronic theory of covalent bonding** had been worked out

A.2 Carbon's tetravalence and the shapes it gives

How carbon compounds take their shape

- a grasp of the **fundamental concepts of molecular structure** is what lets one understand, and predict, **how an organic compound behaves**
- Unit 4 covered the **theories of valency and molecular structure**; carbon's **electronic configuration** and the **hybridisation of its *s* and *p* orbitals** explain both its **tetravalence** and the covalent bonds it forms
- the formation and the shapes of **methane (CH_4), ethene (C_2H_4) and ethyne (C_2H_2)** follow from carbon using **sp^3, sp^2 and *sp* hybrid orbitals** in each case

- **bond length and bond enthalpy, that is bond strength, both turn on the hybridisation**
- with **more *s* character** the ***sp* hybrid orbital sits nearer the nucleus**, and the bonds it makes are **shorter and stronger** than the sp^3 orbital's
- the **sp^2 orbital falls between *sp* and sp^3** in *s* character, and so do the **length and the enthalpy of the bonds it makes**

- **carbon's electronegativity shifts too** as the hybridisation changes
- the **more *s* character the hybrid orbital carries, the more electronegative that carbon**
- a carbon with an ***sp* hybrid orbital, 50% *s* character**, is **more electronegative** than one holding sp^2 or sp^3 orbitals
- that difference shows up in **a number of the physical and chemical properties** of the molecules in question

Some characteristic features of π bonds

- forming a **π (pi) bond** needs the two ***p* orbitals on adjacent atoms to lie parallel**, for a proper **sideways overlap**
- so **one plane must hold every atom of $H_2C{=}CH_2$**: the two *p* orbitals **run parallel to one another** and **stand at right angles to that plane**
- rotating one CH_2 fragment against the other **interferes with maximum overlap of the *p* orbitals**, so **rotation about a C=C double bond is restricted**
- the π bond's **electron charge cloud sits above and below the plane of the bonding atoms**, which leaves those **electrons easily available to an attacking reagent**
- as a rule, the **most reactive centre in a molecule with multiple bonds is the π bond**

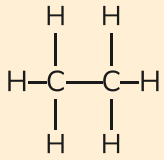
Worked examples 8.a–8.c

- **8.a** How many σ and π bonds are present in (a) $HC{\equiv}CCH{=}CHCH_3$ and (b) $CH_2{=}C{=}CHCH_3$?
- (a) σ_{C-C}: 4; σ_{C-H}: 6; $\pi_{C=C}$: 1; π C≡C: 2 · (b) σ_{C-C}: 3; σ_{C-H}: 6; $\pi_{C=C}$: 2
- **8.b** Name the hybridisation of every carbon in (a) CH_3Cl, (b) $(CH_3)_2CO$, (c) CH_3CN, (d) $HCONH_2$, (e) $CH_3CH{=}CHCN$?
- (a) sp^3; (b) sp^3, sp^2; (c) sp^3, *sp*; (d) sp^2; (e) sp^3, sp^2, sp^2, *sp*
- **8.c** For (a) $H_2C{=}O$, (b) CH_3F, (c) $HC{\equiv}N$, give the carbon's hybridisation and the molecular shape.
- (a) sp^2, trigonal planar; (b) sp^3, tetrahedral; (c) *sp*, linear

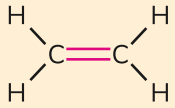
A.3 Putting organic structures on paper

Complete, condensed and bond-line formulas

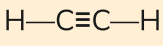
- there is more than one way to write a structure: the **Lewis or dot structure**, the **dash structure**, the **condensed structure**, and the **bond line structural formula**
- draw the **two-electron covalent bond as a dash (–)** and the Lewis structure simplifies, the attention falling on the **electrons that form the bond**
- a **single dash is a single bond**, a **double dash a double bond**, a **triple dash a triple bond**
- **lone pairs on a heteroatom** — oxygen, nitrogen, sulphur, a halogen — **may or may not be shown**
- structural representations of that kind are called **complete structural formulas**



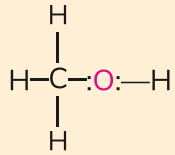
Ethane



Ethene



Ethyne



Methanol

- those formulas **abbreviate further** when some or all of the dashes are omitted and **identical groups on one atom are counted by a subscript**
- the expression that results is a **condensed structural formula**

CH_3CH_3	$H_2C{=}CH_2$	$HC{\equiv}CH$	CH_3OH
Ethane	*Ethene*	*Ethyne*	*Methanol*

- so **$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_3$** condenses to **$CH_3(CH_2)_6CH_3$**

- for further simplification organic chemists use a representation made **of lines alone**, the **bond-line structural formula**
- there **no carbon or hydrogen atom is written in**, and the **bonds between carbons are drawn as a zig-zag**
- the only atoms written in are **oxygen, chlorine, nitrogen** and the like
- the **terminals stand for methyl (–CH_3) groups**, unless a functional group says otherwise
- each **line junction is a carbon carrying as many hydrogens as its valency needs**

(i) 3-Methyloctane

(a) $CH_3CH_2CH(CH_3)CH_2CH_2CH_2CH_2CH_3$

(b) $CH_3–CH_2–CH(CH_3)–CH_2–CH_2–CH_2–CH_2–CH_3$ (c)

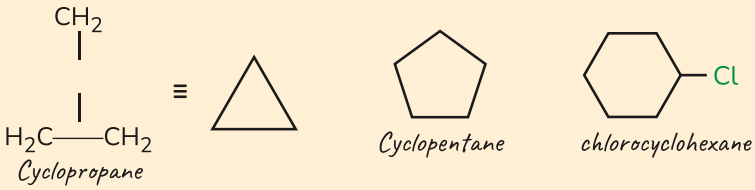


(ii) 2-Bromobutane

(a) $CH_3CHBrCH_2CH_3$ (b) $CH_3–CH(Br)–CH_2–CH_3$ (c)



- in a **cyclic compound** the bond-line formula is drawn as the ring itself

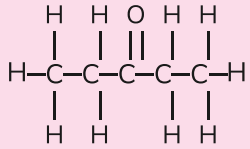


- the ring drawn plain carries a **CH_2 at every unlabelled corner**; only a substituent is written in

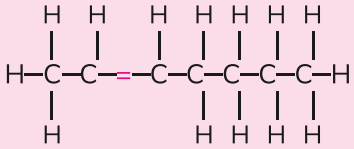
Worked example 8.d

Expand each condensed formula into its complete structural formula.

(a) $CH_3CH_2COCH_2CH_3$ ⟶



(b) $CH_3CH{=}CH(CH_2)_3CH_3$ ⟶



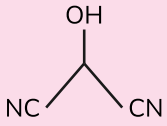
Worked example 8.e

For each compound, write a condensed formula and a bond-line formula.

(a) $HOCH_2CH_2CH_2CH(CH_3)CH(CH_3)CH_3$ → $HO(CH_2)_3CH(CH_3)CH(CH_3)_2$ →



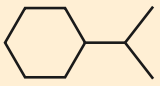
(b) $N{\equiv}C{-}CH(OH){-}C{\equiv}N$ → $HOCH(CN)_2$ →



Worked example 8.f

Expand each bond-line formula to show every atom, carbon and hydrogen alike.

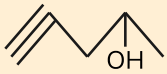
(a)



(b)



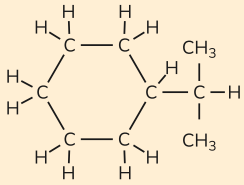
(c)



(d)

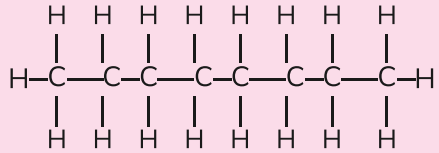


Solution (a):

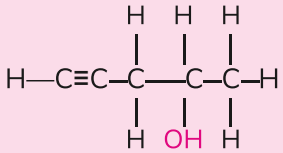


Worked example 8.f · the rest of the answer

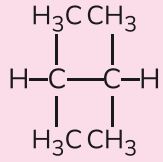
(b)



(c)



(d)



Three-dimensional representation of organic molecules

- settled conventions put an organic molecule's **three-dimensional (3-D) structure** onto paper
- a **solid wedge** and a **dashed wedge** together give the **wedge formula**, from which the 3-D image can be read off a two-dimensional picture
- the **solid-wedge** stands for a bond that **comes up out of the paper, towards the reader**
- the **dashed-wedge** marks a bond **projecting out of the plane and away from the observer**
- a wedge is drawn with its **broad end towards the observer**
- bonds **lying in the plane of the paper** are drawn as a **normal line (—)**
- Figure 8.A puts methane on paper this way

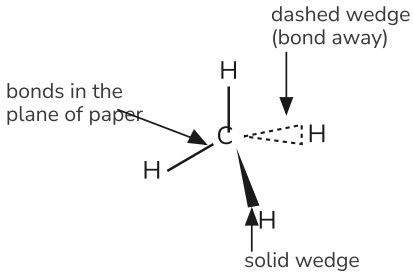


Figure 8.A Wedge-and-dash representation of CH_4

Molecular Models

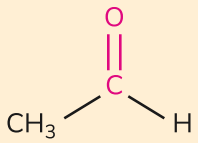
- a molecular model is a **physical device** for **seeing and grasping an organic molecule's three-dimensional shape**; models are made of **wood, of plastic or of metal**, and are sold commercially
- three types are in common use: the **Framework model**, the **Ball-and-stick model** and the **Space filling model**
- the **framework model** shows **only the bonds connecting the atoms, not the atoms themselves**: it emphasises the **pattern of bonds** and ignores the **size of the atoms**
- the **ball-and-stick model** shows **both**: **balls are atoms, a stick is a bond**; where a compound holds a C=C, as ethene does, **springs stand in for the sticks**, making it a **ball-and-spring model**
- the **space-filling model** puts the weight on **how large each atom is beside the others, going by the van der Waals radius**; **no bond appears**, and what it conveys is the **volume each atom takes up** in the molecule
- besides these, **computer graphics** can be used for molecular modelling

Figure 8.B Fig. 8.2

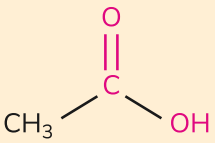
- the **framework model** of methane shows four sticks meeting at a point, with nothing drawn where the atoms are
- the **ball and stick model** puts a large dark ball at the carbon and four smaller pale balls at the hydrogens, the sticks between them the bonds
- the **space filling model** shows the same atoms swollen until they touch, so the sticks disappear and only the **volume the molecule occupies** is left

A.4 Classification of organic compounds

- organic compounds are **so many, and still growing in number**, that they must be **classified by their structures**
- the broad classification runs: **I · Acyclic or open chain compounds**
- **II · Cyclic or closed chain, that is ring, compounds**, which divide into
  - **alicyclic compounds** — **homocyclic or carbocyclic**, and **heterocyclic (non-aromatic)**
  - **aromatic compounds** — **benzenoid** and **non-benzenoid**, and the **heterocyclic aromatic** ones



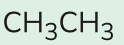
Acetaldehyde



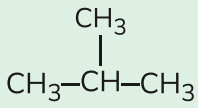
Acetic acid

I · Acyclic or open chain compounds

- these are also called **aliphatic** compounds, and they consist of **straight or branched chain** compounds



Ethane



Isobutane

II · Cyclic or closed chain compounds · (a) Alicyclic compounds

- an **alicyclic (aliphatic cyclic)** compound holds **carbon atoms joined in a ring**, which makes it **homocyclic**


Cyclopropane


Cyclohexane


Cyclohexene

- sometimes **atoms other than carbon are present in the ring**, which makes the compound **heterocyclic**; **tetrahydrofuran** is one such
- these show **some of the properties of aliphatic compounds**



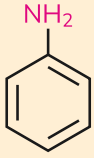
Tetrahydrofuran

(b) Aromatic compounds

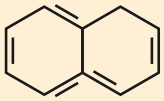
- a special class, taken up in detail in Unit 9; they include **benzene and other related ring compounds**, the **benzenoid** ones
- as alicyclic compounds may, an aromatic compound may also carry a **hetero atom in the ring**, and is then a **heterocyclic aromatic** compound

Benzenoid aromatic —

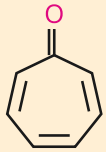

Benzene



Aniline


Naphthalene

Non-benzenoid —



Tropone

Heterocyclic aromatic compounds



Functional group

- organic compounds may also be classified by their **functional groups**, into **families** or **homologous series**
- a **functional group** is the **atom, or group of atoms, attached to the carbon chain**, and the compound's **characteristic chemical properties answer to it**
- the **hydroxyl group (–OH)**, the **aldehyde group (–CHO)** and the **carboxylic acid group (–COOH)** are examples

Homologous series

- organic compounds that all carry one **characteristic functional group** make up a **homologous series**, its members being **homologues**
- the members answer to one **general molecular formula**, and successive members **differ by a $–CH_2$ unit**
- the **alkanes and alkenes, the alkynes, the haloalkanes, the alkanols and alkanals, the alkanones, the alkanoic acids and the amines** are all such series
- a compound may hold **two or more identical or different functional groups**, which makes it **polyfunctional**

A.5 Nomenclature of organic compounds

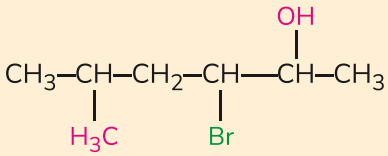
- with **millions of compounds** to deal with, organic chemistry had to develop a **systematic way of naming** them, so that each could be identified without ambiguity
- that system belongs to the **International Union of Pure and Applied Chemistry, IUPAC**
- in it the **name answers to the structure**, so that anyone who hears the name can **work the structure out from it**
- before IUPAC, a compound took its name from **where it came from** or from some **property** of it: the acid of **citrus fruit** is **citric acid**, and the acid of the **red ant**, Latin *formica*, is **formic acid**
- such traditional names count as **trivial or common names**, and some are followed even today
- the newly discovered **C_{60} cluster**, a form of carbon, was given the common name **Buckminsterfullerene**: its structure resembles the **geodesic domes** the architect **R. Buckminster Fuller** made popular
- common names remain useful, and in many cases indispensable, above all where the **systematic name runs long and complicated**

TABLE 8.A Common or Trivial Names of Some Organic Compounds

Compound	Common name
CH_4	Methane
$H_3CCH_2CH_2CH_3$	*n*-Butane
$(H_3C)_2CHCH_3$	Isobutane
$(H_3C)_4C$	Neopentane
$H_3CCH_2CH_2OH$	*n*-Propyl alcohol
$HCHO$	Formaldehyde
$(H_3C)_2CO$	Acetone
$CHCl_3$	Chloroform
CH_3COOH	Acetic acid
C_6H_6	Benzene
$C_6H_5OCH_3$	Anisole
$C_6H_5NH_2$	Aniline
$C_6H_5COCH_3$	Acetophenone
$CH_3OCH_2CH_3$	Ethyl methyl ether

The IUPAC system of nomenclature

- to build a systematic name, pick out the **parent hydrocarbon** first, and then **whatever functional groups hang on it**



the chain is the parent; H_3C is a branch; OH and Br are functional groups

- **prefixes and suffixes** modify the parent name to give the actual one
- a compound of **carbon and hydrogen only** is a **hydrocarbon**
- one holding **nothing but C–C single bonds** is **saturated**; IUPAC calls that homologous series the **alkanes**
- their older name, **Paraffin**, is Latin for **little affinity**
- an **unsaturated hydrocarbon** has **one C=C or C≡C at the least**

IUPAC nomenclature of alkanes

Naming a straight chain

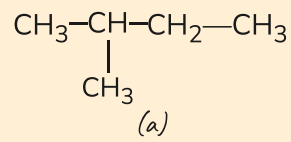
- their names rest on the **chain structure**, end in the suffix **'-ane'**, and carry a **prefix giving the number of carbon atoms in the chain**
- the exception runs from **CH_4 to C_4H_{10}**, whose names come from **trivial names**
- the alkanes of Table 8.B **differ only by the number of -CH_2 groups** in the chain, so they are **homologues** of the alkane series

TABLE 8.B **IUPAC Names of Some Unbranched Saturated Hydrocarbons**

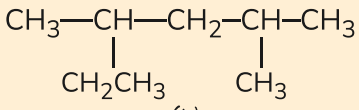
Name	Molecular formula	Name	Molecular formula
Methane	CH_4	Heptane	C_7H_{16}
Ethane	C_2H_6	Octane	C_8H_{18}
Propane	C_3H_8	Nonane	C_9H_{20}
Butane	C_4H_{10}	Decane	$C_{10}H_{22}$
Pentane	C_5H_{12}	Icosane	$C_{20}H_{42}$
Hexane	C_6H_{14}	Triacontane	$C_{30}H_{62}$

Naming a branched chain

- a branched compound has **short carbon chains hanging off one or more carbons of the parent chain**
- those small chains, the branches, are the **alkyl groups**



(a)



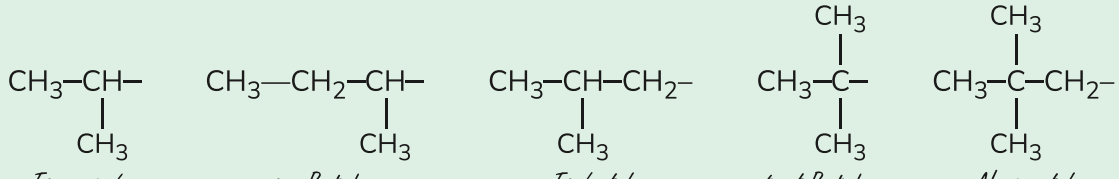
(b)

- to name such compounds, the **names of the alkyl groups are prefixed to the name of the parent alkane**
- take **one hydrogen off a carbon** of a saturated hydrocarbon and what is left is an **alkyl group**: CH_4 gives -CH_3, **methyl**
- an alkyl group is named by **putting 'yl' for 'ane'** in the corresponding alkane

TABLE 8.C **Some Alkyl Groups**

Molecular formula	Name of alkane	Structural formula	Name of alkyl group
CH_4	Methane	$-CH_3$	Methyl
C_2H_6	Ethane	$-CH_2CH_3$	Ethyl
C_3H_8	Propane	$-CH_2CH_2CH_3$	Propyl
C_4H_{10}	Butane	$-CH_2CH_2CH_2CH_3$	Butyl
$C_{10}H_{22}$	Decane	$-CH_2(CH_2)_8CH_3$	Decyl

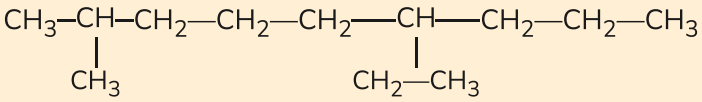
- some alkyl groups are **abbreviated**: methyl as **Me**, ethyl as **Et**, propyl as **Pr** and butyl as **Bu**
- alkyl groups may themselves be **branched**, so propyl and butyl groups have branched forms too
- branched groups have their own **trivial names**: a propyl group is either ***n*-propyl** or **isopropyl**, and the branched butyl groups are ***sec*-butyl, isobutyl and *tert*-butyl**
- the structural unit **$-CH_2C(CH_3)_3$** is the **neopentyl** group



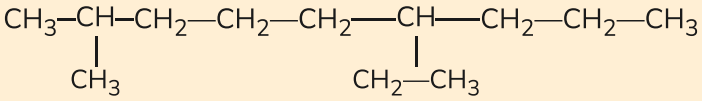
Isopropyl- *sec-Butyl-* *Isobutyl-* *tert-Butyl-* *Neopentyl-*

Nomenclature of branched chain alkanes · the rules

- **1** first identify the **longest carbon chain in the molecule**. In the case drawn here that chain runs to **nine carbons**, and it is the **parent**, or root, chain. The selection shown as **II is not correct**, having **only eight carbons**.

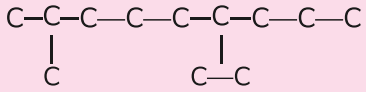


I · nine carbons in the parent chain



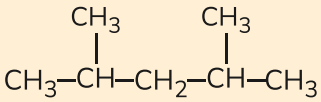
II · counting only eight carbons is wrong

- **2 number the carbon atoms of the parent chain**, which fixes the parent alkane and pins down **where the branches sit**, each branch being an alkyl group in place of a hydrogen
- in the example above, number **from left to right**, which puts the branches at **carbons 2 and 6**, not from right to left, which would give them **4 and 8**
- **3** the **alkyl group names go in front of the parent alkane's name**, each with **a number saying where it sits**
- several **unlike alkyl groups** go in **alphabetical order**
- so the compound above is **6-ethyl-2-methylnonane**. A **hyphen** sits between each number and its group, and **methyl runs straight into nonane**.

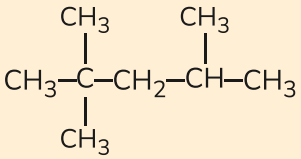


1 to 9 from the left puts the branches at 2 and 6

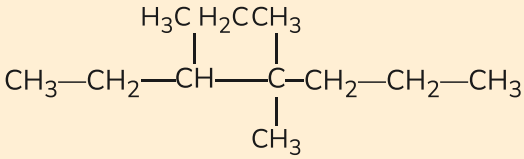
- **4** when the **same substituent appears twice or more**, its **locants are set off by commas**
- the name of the substituent is **not repeated**; instead the prefixes **di, tri, tetra, penta, hexa** (2, 3, 4, 5, 6) and so on are used
- in **alphabetical listing**, however, **those prefixes are not counted**



2,4-Dimethylpentane

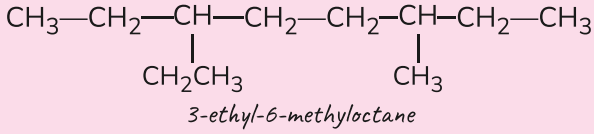


2,2,4-Trimethylpentane

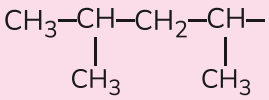


3-Ethyl-4,4-dimethylheptane

- **5** where the **two substituents stand in equivalent positions**, the **alphabetically earlier one takes the lower number**
- so the compound below is **3-ethyl-6-methyloctane**, not 6-ethyl-3-methyloctane
- **6** a **branched alkyl group** is named by the same procedures, except that **numbering starts at the branch carbon joined to the root alkane**

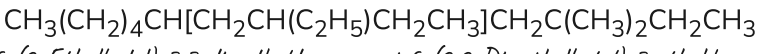


3-ethyl-6-methyloctane



1,3-Dimethylbutyl- · numbered 1 at the point of attachment

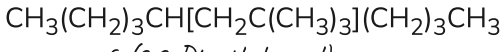
- the name of a branched chain alkyl group is put **in parenthesis** when the compound is named
- when substituents' trivial names are **alphabetised**, the prefixes **iso- and neo- count as part of the fundamental name** of the alkyl group, while ***sec-* and *tert-* do not**
- in multi-substituted compounds: where **two chains of equal size** offer themselves, take the one carrying **more side chains**; and once the chain is chosen, **number from the end closer to the substituent**



5-(2-Ethylbutyl)-3,3-dimethyldecane, not 5-(2,2-Dimethylbutyl)-3-ethyldecane



5-sec-Butyl-4-isopropyldecane



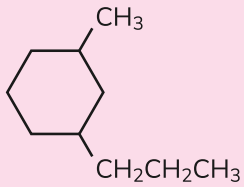
5-(2,2-Dimethylpropyl)nonane

Rings

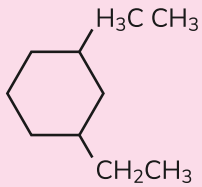
- a **saturated single ring** takes the **straight chain alkane's name with 'cyclo' in front**
- where **side chains** are present, the rules above apply
- numbering goes **alphabetically**; where a choice remains, the **more branched carbon takes the lower number**



Cyclopentane



1-Methyl-3-propylcyclohexane

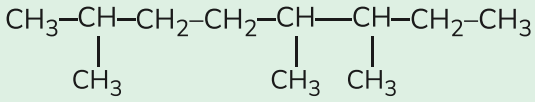


3-Ethyl-1,1-dimethylcyclohexane and not 1-ethyl-3,3-dimethylcyclohexane

Worked example 8.g

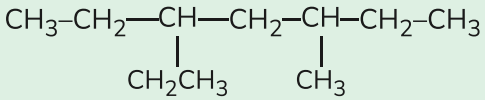
Why is each name in parentheses incorrect?

(a)



2,5,6-Trimethyloctane and not 3,4,7-Trimethyloctane

(b)



3-Ethyl-5-methylheptane and not 5-Ethyl-3-methylheptane

- **(a)** the **lowest locant set wins**: 2,5,6 is lower than 3,5,7
- **(b)** the substituents stand in **equivalent positions**, so the **lower number goes to the one coming first alphabetically**

Nomenclature of organic compounds having functional group(s)

- a **functional group**, defined earlier, is an **atom or a cluster of atoms bonded in a particular way**, and it is usually the **site of a chemical reaction** — continued on sheet 11

- a functional group is usually the **site of chemical reactivity** in an organic molecule
- compounds carrying the **same functional group undergo similar reactions**: CH_3OH, CH_3CH_2OH and $(CH_3)_2CHOH$ all hold -OH, and each **gives off hydrogen with sodium metal**
- functional groups are what let organic compounds be **sorted systematically into classes**
- Table 8.D sets out a number of functional groups, each with its prefix, its suffix and an example compound

- first, **identify the functional group** present, which settles the **choice of suffix**
- **take the longest carbon chain that holds the group, and number it so the group lands on the lowest carbon it can**
- the suffix from Table 8.D then gives the name

- in a **polyfunctional compound** one group is picked as the **principal functional group**, and the name rests on it
- the remaining groups are **subordinate functional groups**, named as **substituents with the appropriate prefixes**
- which group is principal follows a **ranking**; for some common groups the **priority falls in this order**
- **-COOH, $–SO_3H$, -COOR (R = alkyl group), COCl, $-CONH_2$, -CN, -HC=O, >C=O, -OH, $-NH_2$, >C=C<, -C≡C-**
- the groups **–R, C_6H_5-, the halogens (F, Cl, Br, I), $–NO_2$, alkoxy (–OR)** and the like are **always prefix substituents**
- so a compound holding **both an alcohol and a keto group is named as a hydroxyalkanone**, the keto group being preferred to the hydroxyl
- $HOCH_2(CH_2)_3CH_2COCH_3$ is **7-hydroxyheptan-2-one**, not 2-oxoheptan-7-ol; likewise $BrCH_2CH{=}CH_2$ is **3-bromoprop-1-ene**, not 1-bromoprop-2-ene
- with **two or more groups of one kind**, their number is shown by adding **di, tri** and so on before the class suffix; the **full name of the parent alkane is then written before that suffix**
- $CH_2(OH)CH_2(OH)$ is named **ethane–1,2–diol**
- the **ending –ne of the parent alkane is dropped** where the compound holds one double or triple bond: $CH_2{=}CH{-}CH{=}CH_2$ is **buta–1,3–diene**

Worked example 8.h

Give the IUPAC name of each structure.

(i) $CH_3–CH_2–\underset{\displaystyle OH}{\underset{|}{CH}}–CH_2–CH_2–\underset{\displaystyle CH_3}{\underset{|}{CH}}–CH_2–CH_3$

- the functional group is an **alcohol (OH)**, so the suffix is **'-ol'**
- the **longest chain containing the -OH has eight carbon atoms**, so the saturated hydrocarbon is **octane**
- the **-OH is on carbon atom 3**, and a **methyl group is attached at the 6th carbon**
- so the systematic name is **6-Methyloctan-3-ol**

(ii) $^6CH_3–^5CH_2–{}^4\overset{\displaystyle O}{\overset{\|}{C}}–^3CH_2–{}^2\overset{\displaystyle O}{\overset{\|}{C}}–^1CH_3$

- the functional group is a **ketone (>C=O)**, so the suffix is **'-one'**; **two keto groups** are shown by **'di'**, making the suffix **'dione'**
- the keto groups are at **carbons 2 and 4**, and the longest chain holds **6 carbon atoms**, so the parent hydrocarbon is **hexane**
- the systematic name is **Hexane-2,4-dione**

(iii) $^6CH_3–{}^5\overset{\displaystyle O}{\overset{\|}{C}}–^4CH_2–^3CH_2–^2CH_2–^1COOH$

TABLE 8.D Some Functional Groups and Classes of Organic Compounds

Class of compounds	Functional group structure	IUPAC group prefix	IUPAC group suffix	Example
Alkanes	-	-	-ane	Butane, $CH_3(CH_2)_2CH_3$
Alkenes	>C=C<	-	-ene	But-1-ene, $CH_2=CHCH_2CH_3$
Alkynes	-C≡C-	-	-yne	But-1-yne, $CH{\equiv}CCH_2CH_3$
Arenes	-	-	-	Benzene, 
Halides	-X (X=F,Cl,Br,I)	halo-	-	1-Bromobutane, $CH_3(CH_2)_2CH_2Br$
Alcohols	–OH	hydroxy-	-ol	Butan-2-ol, $CH_3CH_2CHOHCH_3$
Aldehydes	–CHO	formyl, or oxo	-al	Butanal, $CH_3(CH_2)_2CHO$
Ketones	>C=O	oxo-	-one	Butan-2-one, $CH_3CH_2COCH_3$
Nitriles	-C≡N	cyano	nitrile	Pentanenitrile, $CH_3CH_2CH_2CH_2CN$
Ethers	–R–O–R–	alkoxy-	-	Ethoxyethane, $CH_3CH_2OCH_2CH_3$
Carboxylic acids	–COOH	carboxy	-oic acid	Butanoic acid, $CH_3(CH_2)_2CO_2H$
Carboxylate ions	$–COO^-$	-	-oate	Sodium butanoate, $CH_3(CH_2)_2CO_2^-\ Na^+$
Esters	–COOR	alkoxycarbonyl	-oate	Methyl propanoate, $CH_3CH_2COOCH_3$
Acyl halides	–COX (X=F,Cl,Br,I)	halocarbonyl	-oyl halide	Butanoyl chloride, $CH_3(CH_2)_2COCl$
Amines	$–NH_2$, >NH, >N-	amino-	-amine	Butan-2-amine, $CH_3CHNH_2CH_2CH_3$
Amides	$–CONH_2$, -CONHR, $-CONR_2$	-carbamoyl	-amide	Butanamide, $CH_3(CH_2)_2CONH_2$
Nitro compounds	$–NO_2$	nitro	-	1-Nitrobutane, $CH_3(CH_2)_3NO_2$
Sulphonic acids	$–SO_3H$	sulpho	sulphonic acid	Methylsulphonic acid, CH_3SO_3H

Worked example 8.h · continued

- **(iii)** two functional groups are present, a **ketone** and a **carboxylic acid**
- the **acid outranks the ketone**, so the parent chain takes the suffix **'oic acid'**
- numbering **starts at the carbon of the –COOH group**; the **keto group at carbon 5** is shown by **'oxo'**
- the longest chain running through the –COOH carbon holds **6 carbon atoms**, so the parent hydrocarbon is **hexane**, and the compound is **5-Oxohexanoic acid**

(iv) $CH{\equiv}C–CH{=}CH–CH{=}CH_2$

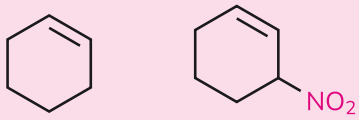
numbered 6 to 1 from the left

- the two **C=C** groups are at carbons **1 and 3**, the **C≡C** at carbon **5**
- those groups take the suffixes **'diene'** and **'yne'**
- the longest chain holding them has **6 carbon atoms**, so the parent hydrocarbon is **hexane** and the name is **Hexa-1,3-dien-5-yne**

Worked example 8.i

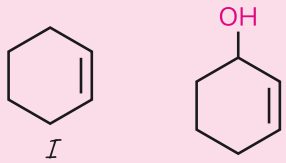
Draw out (i) 2-Chlorohexane, (ii) Pent-4-en-2-ol, (iii) 3-Nitrocyclohexene, (iv) Cyclohex-2-en-1-ol, (v) 6-Hydroxyheptanal.

- **(i)** 'hexane' puts **6 carbon atoms in the chain**, and the functional group **chloro is at carbon 2**, so the structure is **$CH_3CH_2CH_2CH_2CH(Cl)CH_3$**
- **(ii)** 'pent' puts **5 carbon atoms in the chain**; 'en' and 'ol' answer to **C=C at carbon 4 and -OH at carbon 2**, so the structure is **$CH_2{=}CHCH_2CH(OH)CH_3$**
- **(iii)** 'cyclohexene' implies a **six membered ring with a carbon-carbon double bond**, numbered as in (I)
- the prefix **3-nitro** puts a nitro group on **C-3**, which gives (II)
- the **double bond is a suffixed functional group** while **NO_2 is a prefixed** one, so the **double bond takes preference over $–NO_2$** in the numbering



I *II*

- **(iv)** '1-ol' puts an **-OH group at C-1**; OH is a **suffixed functional group and takes preference over the C=C bond**, which gives (II)

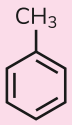


I *II*

- **(v)** 'heptanal' makes the compound an **aldehyde with 7 carbon atoms in the parent chain**, and '6-hydroxy' puts an **-OH at carbon 6**
- so the structural formula is **$CH_3CH(OH)CH_2CH_2CH_2CH_2CHO$**; the **carbon of the –CHO group is counted in the numbering**

Nomenclature of substituted benzene compounds

- for the IUPAC name of a substituted benzene, the **substituent is placed as a prefix to the word benzene**
- many substituted benzenes also go by **common names** that **everyone uses**; sheet 14 has examples



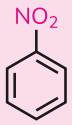
Methylbenzene (Toluene)



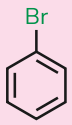
Methoxybenzene (Anisole)



Aminobenzene (Aniline)

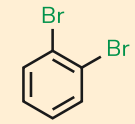


Nitrobenzene

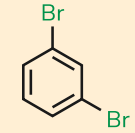


Bromobenzene

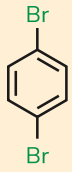
- where a benzene ring is **disubstituted**, the positions of the substituents are fixed by **numbering the ring carbons so that the substituents fall on the lowest numbers possible**
- so compound (b) below is **1,3-dibromobenzene**, not 1,5-dibromobenzene
- the **trivial system** marks the relative positions **1,2 · 1,3 · 1,4** with the prefixes ***ortho* (o), *meta* (m), *para* (p)**
- so 1,3-dibromobenzene (b) is ***m*-dibromobenzene**, and the other isomers, 1,2-(a) and 1,4-(c), are ***ortho*** (or just *o*-) and ***para*** (or just *p*-) dibromobenzene
- for **tri- or higher substituted** benzene derivatives those prefixes **cannot be used**: the compounds are named by **identifying the substituent positions on the ring under the lowest locant rule**
- in some cases the **common name of a benzene derivative is taken as the base compound**



(a) 1,2-Dibromobenzene

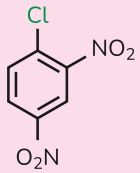


(b) 1,3-Dibromobenzene

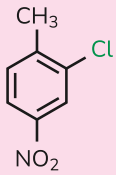


(c) 1,4-Dibromobenzene

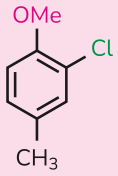
- the **base compound's own substituent takes position 1**, and numbering **runs the way that gives the next substituent the smaller number**
- the substituents then appear in the name in **alphabetical order**
- when a benzene ring is **attached to an alkane carrying a functional group**, the ring counts as a **substituent, not as the parent**
- the name for benzene as a substituent is **phenyl (C_6H_5-)**, abbreviated **Ph**



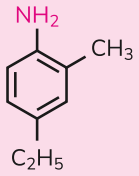
1-Chloro-2,4-dinitrobenzene, not 4-chloro-1,3-dinitrobenzene



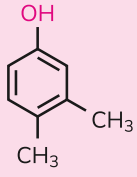
2-Chloro-1-methyl-4-nitrobenzene, not 4-methyl-5-chloro-nitrobenzene



2-Chloro-4-methylanisole



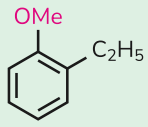
4-Ethyl-2-methylaniline



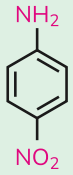
3,4-Dimethylphenol

Worked example 8.j

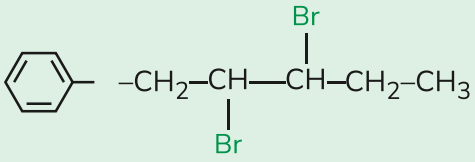
Draw (a) *o*-Ethylanisole, (b) *p*-Nitroaniline, (c) 2,3-Dibromo-1-phenylpentane, (d) 4-Ethyl-1-fluoro-2-nitrobenzene.



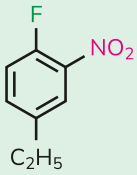
(a)



(b)



(c)



(d)

A.6 Isomerism

- where **two or more compounds share a molecular formula but differ in properties**, the phenomenon is **isomerism**, and the compounds are **isomers**
- isomerism divides as follows

Structural isomerism

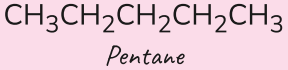
- chain isomerism
- position isomerism
- functional group isomerism
- metamerism

Stereoisomerism

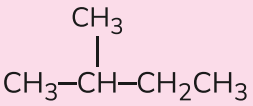
- geometrical isomerism
- optical isomerism

Structural isomerism

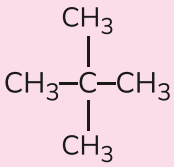
- compounds sharing a **molecular formula but differing in structure**, that is in the way the atoms are linked, are **structural isomers**
- **(i) Chain isomerism** — where two or more compounds of similar molecular formula carry **different carbon skeletons**, they are **chain isomers**
- C_5H_{12} represents **three compounds**



Pentane

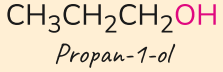


Isopentane (2-Methylbutane)

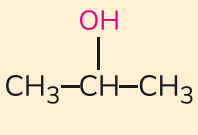


Neopentane (2,2-Dimethylpropane)

- **(ii) Position isomerism** — where two or more compounds differ in **where on the carbon skeleton a substituent atom or functional group sits**, they are **position isomers**
- C_3H_8O represents **two alcohols**

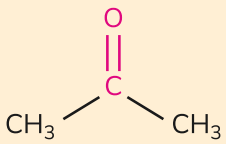


Propan-1-ol

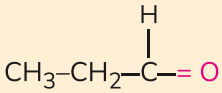


Propan-2-ol

- **(iii) Functional group isomerism** — **functional isomers** share a **molecular formula but carry unlike functional groups**
- C_3H_6O represents **an aldehyde and a ketone** — set out on sheet 16



Propanone



Propanal

- **(iv) Metamerism** — it comes of **unequal alkyl chains flanking the functional group**
- $C_4H_{10}O$ represents **methoxypropane ($CH_3OC_3H_7$)** and **ethoxyethane ($C_2H_5OC_2H_5$)**

Stereoisomerism

- compounds sharing the **same constitution and the same sequence of covalent bonds**, yet with their **atoms or groups set out differently in space**, are **stereoisomers**
- this special kind of isomerism is **stereoisomerism**, and it divides into **geometrical** and **optical** isomerism

B.1 How organic reactions happen: mechanism

- an organic reaction sets the **organic molecule, the substrate**, against a **suitable attacking reagent**; **one or more intermediates** come first and the **product(s)** last
- the **substrate** is the reactant that **supplies carbon to the new bond**; the other reactant is the **reagent**
- where **both reactants supply carbon**, the choice is **arbitrary**, and the molecule attention is focused on is then called the substrate
- along the way a covalent bond **C–C or C–other atom, breaks, and a fresh bond is made**
- the **reaction mechanism** is the **step-by-step account**: where the **electrons move**, the **energetics of each bond broken and made**, and how **fast reactants turn into products (kinetics)**
- with the mechanism in hand one can **make sense of an organic compound's reactivity** and **plan how to synthesise it**

Organic molecule (Substrate)
Attacking Reagent → [Intermediate]
→ Product(s)
with byproducts alongside

Fission of a covalent bond

- a covalent bond can break **two ways**: **(i) heterolytically** or **(ii) homolytically**
- in **heterolysis one fragment keeps the whole shared pair**
- that leaves **one atom with only six electrons and a positive charge**, and the other **with a full octet, a lone pair at the least, and a negative charge**
- so heterolytic cleavage of **bromomethane** gives **$^+CH_3$ and Br^-**



the curved arrow shows the pair of electrons moving to bromine

- a species whose **carbon atom carries a sextet of electrons and a positive charge** is a **carbocation**, earlier called a **carbonium ion**
- **CH_3^+** is the **methyl cation**, once the methyl carbonium ion
- a carbocation is **primary, secondary or tertiary** according as the **charged carbon has one, two or three carbons bonded to it**
- other examples are **$CH_3^+CH_2$** (ethyl cation, primary), **$(CH_3)_2{}^+CH$** (isopropyl cation, secondary) and **$(CH_3)_3C^+$** (*tert*-butyl cation, tertiary)
- a carbocation is **very unstable, and very reactive**
- alkyl groups on the charged carbon **steady it**, through the **inductive and hyperconjugation effects** (sections B.1.5 and B.1.9)
- the observed order of carbocation stability is **$^+CH_3 < CH_3^+CH_2 < (CH_3)_2{}^+CH < (CH_3)_3C^+$**
- these carbocations have a **trigonal planar shape**, the positively charged carbon being **sp^2 hybridised**
- so the shape of $^+CH_3$ may be read as **three equivalent C(sp^2) orbitals each overlapping a hydrogen 1*s***

- each bond may be taken as a **C(sp^2)–H(1s) sigma bond**
- the **remaining carbon orbital stands perpendicular to the molecular plane and holds no electrons** (Figure 8.C)
- heterolysis can also leave **the shared pair on carbon**
- where a group Z attached to carbon **leaves without the electron pair**, the **methyl anion ($H_3C:^-$)** is formed
- a carbon species carrying a **negative charge on carbon** is a **carbanion**
- carbon in a carbanion is generally **sp^3 hybridised**, and its structure is a **distorted tetrahedron** (Figure 8.D)
- carbanions too are **unstable and reactive**
- reactions running through **heterolysis** are **ionic**, **heteropolar** or simply **polar** reactions

$$CH_3 — Z \longrightarrow H_3C:^- + Z^+$$

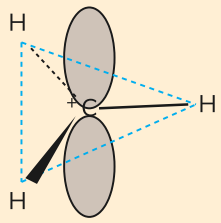


Figure 8.C Shape of methyl carbocation

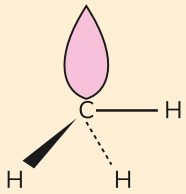


Figure 8.D Shape of methyl carbanion

- in **homolysis** the shared pair **splits, one electron to each atom**
- so the movement is of a **single electron, not of an electron pair**, and a **'half-headed' or fish-hook curved arrow** marks it
- such cleavage gives **neutral species, atom or group, holding an unpaired electron**: these are **free radicals**
- a free radical is **as reactive as a carbocation or carbanion**

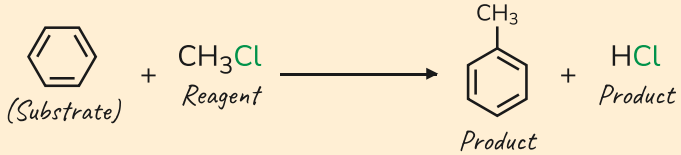
$$R — X \xrightarrow{\text{Heat or Light}} \underset{\textit{Alkyl free radical}}{\dot{R}} + \dot{X}$$

- alkyl radicals are classified **primary, secondary or tertiary**, and their **stability increases from primary to tertiary**
- $\dot{C}H_3 < \dot{C}H_2CH_3 < \dot{C}H(CH_3)_2 < \dot{C}(CH_3)_3$ — methyl, ethyl, isopropyl and *tert*-butyl free radical
- reactions running through **homolysis** are **free radical**, **homopolar** or **nonpolar** reactions

Substrate and reagent

- organic compounds **seldom react as ions**; it is the **whole molecule that takes part**
- it is convenient to name **one reagent as substrate and the other as reagent**
- in general, the molecule whose **carbon is involved in forming the new bond is the substrate**, and the other is the **reagent**
- where a **carbon-carbon bond is formed**, the choice is **arbitrary** and depends on the molecule under observation

(i) $\underset{\textit{Substrate}}{CH_2 = CH_2} + \underset{\textit{Reagent}}{Br_2} \longrightarrow \underset{\textit{Product}}{CH_2Br - CH_2Br}$

(ii) 

Nucleophiles and electrophiles

- a reagent **attacks the reactive site of the substrate**; that site may be an **electron deficient portion**, that is a **positive reactive site** — an atom with an incomplete electron shell, or the positive end of a dipole
- where the **attacking species is electron rich**, it goes for those sites
- an **electron-poor attacker** goes instead for the part of the substrate **able to give electrons up**, a **C=C bond's π electrons** for instance
- a reagent that **arrives carrying an electron pair** is a **nucleophile (Nu:)**, that is **nucleus seeking**, and the reaction is **nucleophilic**
- one that **carries an electron pair off** is an **electrophile (E^+)**, that is **electron seeking**, and the reaction is **electrophilic**
- in a polar reaction the **nucleophile goes for the substrate's electrophilic centre**, the particular electron-poor atom or region
- likewise an **electrophile attacks a nucleophilic centre**, the electron rich part; the electrophile **takes an electron pair off the substrate** and a bond forms between them
- a **curved arrow** traces the pair as it passes from nucleophile to electrophile

Worked example 8.k

Using curved-arrow notation, show how reactive intermediates form when these covalent bonds undergo heterolytic cleavage: (a) CH_3–SCH_3, (b) CH_3–CN, (c) CH_3–Cu.

(a) $CH_3 — SCH_3 \longrightarrow CH_3^+ + {}^-SCH_3$

(b) $CH_3 — CN \longrightarrow CH_3^+ + {}^-CN$

(c) $CH_3 — Cu \longrightarrow {}^-CH_3 + Cu^+$

- typical nucleophiles are **anions carrying a lone pair**: **hydroxide (HO^-)**, **cyanide (NC^-)** and the **carbanions ($R_3C:^-$)**
- **neutral molecules such as $H_2\ddot{O}:$, $R_3N:$, R_2NH** can also act as nucleophiles, for the **lone pair of electrons** they carry
- examples of electrophiles include the **carbocations ($^+CH_3$)** and **neutral molecules holding functional groups** such as the **carbonyl group (>C=O)** or the **alkyl halides (R_3C-X, X a halogen)**
- the carbon of a carbocation has a **sextet configuration**, so it is **short of electrons, and a nucleophile's pair is welcome**
- in a neutral molecule such as an alkyl halide, the **polarity of the C-X bond generates a partial positive charge on carbon**, which makes that carbon an **electrophilic centre a nucleophile can attack**

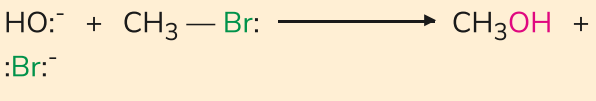
Worked examples 8.l and 8.m

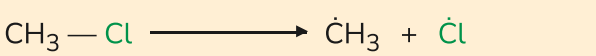
- **8.l** Categorise each of these as nucleophile or electrophile, with your reasons: HS^-, BF_3, $C_2H_5O^-$, $(CH_3)_3N:$, Cl^+, CH_3–C^+=O, $H_2N:^-$, NO_2^+
- **Nucleophiles**: HS^-, $C_2H_5O^-$, $(CH_3)_3N:$, $H_2N:^-$ — each holds an **unshared pair it can give to an electrophile and share**
- **Electrophiles**: BF_3, Cl^+, H_3C–C^+=O, NO_2^+ — their reactive sites hold **only six valence electrons**, and so **can accept an electron pair from a nucleophile**
- **8.m** Identify the electrophilic centre in CH_3CH=O, CH_3CN, CH_3I
- in CH_3H**C***=O, H_3C**C***≡N and H_3**C***–I the starred carbon atoms are the **electrophilic centres**: the **polarity of the bond leaves them with a partial positive charge**

Electron movement in organic reactions

- **curved arrows** are how electron movement is drawn in organic reactions; sheet 19 takes this up

- the notation shows how **changes in bonding follow from electronic redistribution during the reaction**
- to show a **pair of electrons changing position**, the curved arrow **starts where the pair is shifted from and ends where the pair may move to**
- the three cases run: from a **π bond to an adjacent bond position**; from a **π bond to an adjacent atom**; and from an **atom to an adjacent bond position**
- movement of a **single electron** is shown by a **single barbed 'fish hook', that is a half headed curved arrow**





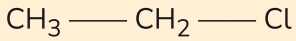
transfer of hydroxide ion giving ethanol, and the dissociation of chloromethane

Electron displacement effects in covalent bonds

- electrons shift within an organic molecule for one of two reasons: **an atom or substituent group pulls on them in the ground state**, or **a suitable attacking reagent comes near**
- a shift owed to **something already in the molecule** makes a **permanent polarisation**
- the **inductive effect** and the **resonance effect** are of this kind
- **temporary** electron displacement effects are seen when a **reagent approaches to attack**; that kind is called the **electromeric effect** or the **polarisability effect**

Inductive effect

- join **two atoms of unequal electronegativity** covalently and the **electron density leans towards the more electronegative one**
- that shift makes the bond a **polar covalent bond**, and **several electronic effects in organic compounds trace back to that polarity**
- take **chlorethane (CH_3CH_2Cl)**, in which the **C–Cl bond is polar**: **carbon-1 turns a little positive (δ^+)**, the **chlorine a little negative (δ^-)**
- those **partial charges** are written **δ (delta)**, and an **arrow from the δ^+ end to the δ^- end** marks which way the density has gone



carbon 2, then carbon 1, then chlorine

$\delta\delta^+$ on carbon 2, δ^+ on carbon 1, δ^- on chlorine

- carbon-1, having developed a **partial positive charge (δ^+)**, in turn **draws electron density towards itself from the adjacent C-C bond**
- so **some positive charge ($\delta\delta^+$) develops on carbon-2** as well, $\delta\delta^+$ standing for a **relatively smaller positive charge** than that on carbon-1
- put another way, the **polar C–Cl bond makes its neighbours polar too**
- such **polarisation of a σ-bond, caused by the polarisation of an adjacent σ-bond, is the inductive effect**
- the effect is **passed on to the subsequent bonds** too, but it **decreases rapidly as the number of intervening bonds rises**, and **three bonds out it has all but gone**

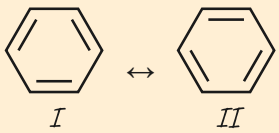
- the effect measures how far a substituent can **withdraw or donate electron density to the carbon it is attached to**
- on that basis substituents are classified, **relative to hydrogen**, as **electron-withdrawing** or **electron donating** groups
- **electron-withdrawing**: the halogens, and many other groups — **$-NO_2$ (nitro), -CN (cyano), -COOH (carboxy), COOR (ester), -OAr (aryloxy, e.g. $-OC_6H_5$)** and the like
- **electron donating**: the alkyl groups, **methyl ($–CH_3$)** and **ethyl ($–CH_2–CH_3$)** among them, are usually so considered

Worked examples 8.n and 8.o

- **8.n** Which bond is more polar in each pair? (a) H_3C-H, H_3C-Br; (b) H_3C-NH_2, H_3C-OH; (c) H_3C-OH, H_3C-SH
- (a) **C–Br**, bromine being more electronegative than hydrogen; (b) **C–O**; (c) **C–O**
- **8.o** In which C–C bond of $CH_3CH_2CH_2Br$ is the inductive effect expected to be the least?
- the **effect weakens with every bond it has to cross**, so the **C-3 to hydrogen bond feels it least**

Resonance structure

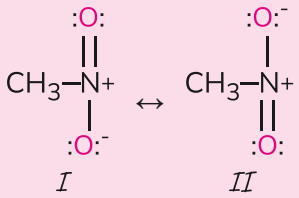
- many organic molecules **behave in ways a single Lewis structure cannot explain**; **benzene** is one
- its **cyclic structure with alternating C–C single and C=C double bonds** is **inadequate for explaining its characteristic properties**
- by that representation benzene should show **two different bond lengths**, one for C–C and one for C=C
- experiment finds instead a **uniform C–C bond distance of 139 pm**, a value **intermediate between the C–C single (154 pm) and the C=C double (134 pm) bond**
- so that structure **does not represent benzene adequately**
- besides, **structures I and II are identical in energy**, and **neither has a better claim** to be benzene



I *II*

- so by **resonance theory** (Unit 4) the actual structure of benzene **cannot be adequately represented by either**: it is a **hybrid of the two**, which are called **resonance structures**
- a **resonance structure**, also called a **canonical or contributing structure**, is a **hypothetical thing: on its own it is no real molecule**
- the **more stable a contributor, the larger its share in the actual structure**

- **nitromethane (CH_3NO_2)** gives another example: it may be written as **two Lewis structures, I and II**, and those structures show **two types of N-O bond**



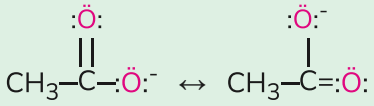
I *II*

- **the two N–O bonds of nitromethane are in fact of the same length, intermediate between an N–O single bond and an N=O double bond**
- so **nitromethane's real structure is a resonance hybrid, I and II blended**

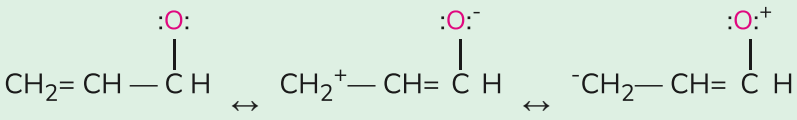
- the **hybrid lies lower in energy than any one canonical structure**
- the **gap between the hybrid and the lowest-lying contributor** is the **resonance stabilisation energy**, or just **resonance energy**
- the **more numerous the important contributing structures, the greater the resonance energy**
- resonance counts for most when the **contributors are of equal energy**
- a set of resonance structures must agree on **(i) where the nuclei sit** and **(ii) how many electrons are unpaired**
- among the resonance structures, the **more stable** is the one with **more covalent bonds**, with **all the atoms carrying an octet of electrons**, hydrogen excepted with its duplet, with **less separation of opposite charges**, with any **negative charge on a more electronegative atom and any positive charge on a more electropositive one**, and with **more dispersal of charge**

Worked examples 8.p – 8.r

- **8.p** Give the resonance structures of CH_3COO^-, tracing the electron movement with curved arrows.
- write the structure, **place each lone pair on its atom**, then **add one arrow at a time**, each moving electrons towards the next structure

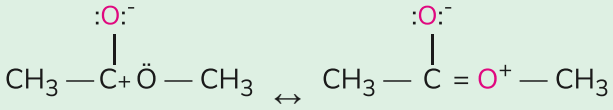


- **8.q** Write the resonance structures of CH_2=CH–CHO, and say which contributing structure is the more stable.



I *II* *III*

- stability runs **I > II > III**
- **I** is the most stable: **more covalent bonds, each carbon and oxygen holding an octet**, and **no separation of opposite charge**
- **II** puts the **negative charge on the more electronegative atom and the positive charge on the more electropositive one**
- **III** does not contribute: **oxygen carries the positive charge and carbon the negative**, which makes it the least stable
- **8.r** Why can neither I nor II below be a major contributor to CH_3COOCH_3?

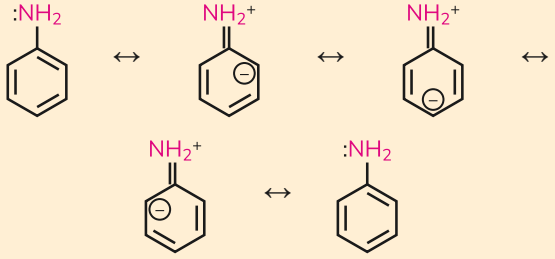


I *II*

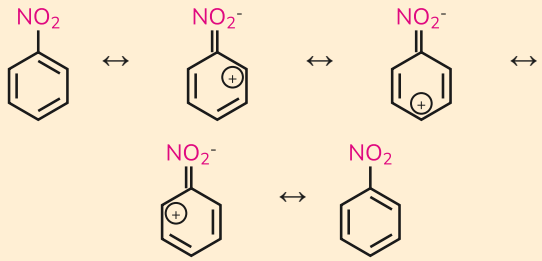
- both are **less important contributors, involving charge separation**; and **in I one carbon lacks a full octet**

Resonance effect

- the **resonance effect** is the **polarity a molecule acquires when two π-bonds interact, or when a π-bond interacts with a lone pair on the next atom**
- it **passes down the chain**
- there are two kinds of **resonance, or mesomeric, effect**, written **R** or **M**
- **(i) Positive resonance effect (+R effect)** — the **electrons move away from the attached atom or group, into the conjugated system**
- that displacement leaves **certain positions in the molecule with high electron density**, as in **aniline**



- **(ii) Negative resonance effect (-R effect)** — here the **electrons move out of the conjugated system, towards the attached atom or group**
- in **nitrobenzene** the displacement runs as below

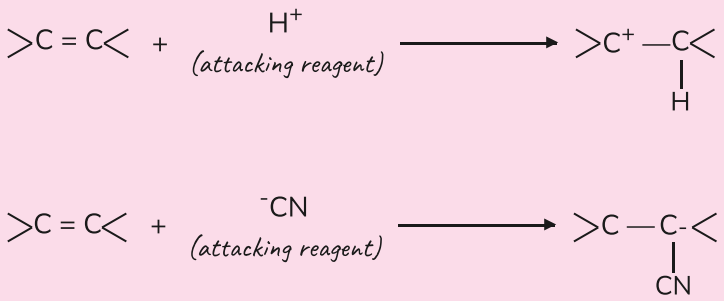


- the atoms or substituent groups showing **+R** electron displacement: – **halogen, –OH, –OR, –OCOR, –NH_2, –NHR, –NR_2, –NHCOR**
- those showing – **R**: – **COOH, –CHO, >C=O, – CN, –NO_2**

- a chain or ring in which **single and double bonds alternate** is a **conjugated system**
- such systems often show **abnormal behaviour**: **1,3-butadiene, aniline and nitrobenzene** are examples
- their **π-electrons spread out, and the system becomes polar**

Electromeric effect (E effect)

- this is a **temporary effect**, shown by organic compounds holding a **multiple bond**, a double or a triple one, and **only in the presence of an attacking reagent**
- it is the **whole π pair of a multiple bond passing to one of its two atoms, because an attacking reagent calls for it**
- take the reagent away and the **effect vanishes**
- its symbol is **E**; a **curved arrow** marks the shift
- **(i) Positive electromeric effect (+E effect)** — the **π pair goes to the atom the reagent will bond to**
- **(ii) Negative electromeric effect (–E effect)** — the **π-electrons are transferred to the atom the attacking reagent does not get attached to**
- where the **inductive and electromeric effects run in opposite directions, the electromeric effect predominates**



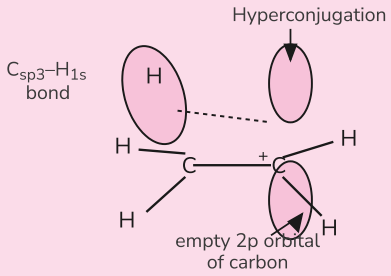
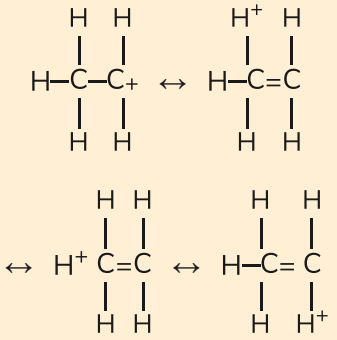


Figure 8.E Orbital diagram showing hyperconjugation in ethyl cation

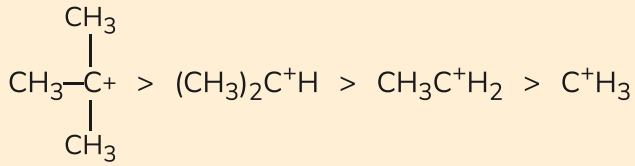
Hyperconjugation

- hyperconjugation is a **general stabilising interaction**: it involves the **delocalisation of the σ electrons of a C—H bond of an alkyl group directly attached to an atom of an unsaturated system, or to an atom carrying an unshared *p* orbital**
- those **σ electrons come partly into conjugation with the unsaturated system, or with the vacant *p* orbital**
- take **$CH_3^+CH_2$, the ethyl cation**, whose positively charged carbon holds an **empty *p* orbital**
- a **methyl C-H bond can line up with that empty *p* orbital**, and its electrons then **spill over into it** (Figure 8.E)
- the overlap **stabilises the carbocation**, electron density from the adjacent σ bond helping to **disperse the positive charge**

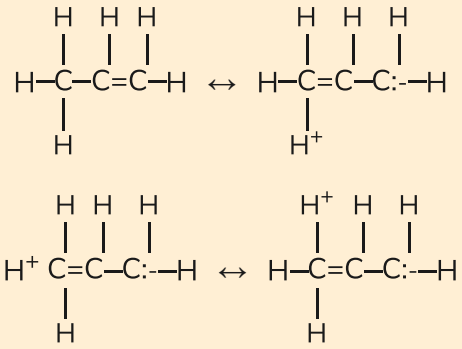


- in general, the **more alkyl groups attached to a positively charged carbon, the greater the hyperconjugation and the greater the stabilisation of the cation**

- so the relative stability of carbocations runs:



- hyperconjugation is **possible in alkenes and alkylarenes** too
- delocalisation of electrons by hyperconjugation in an **alkene** is drawn in Figure 8.F
- hyperconjugation can be viewed several ways; one is to **regard the C—H bond as possessing partial ionic character, through resonance**
- hyperconjugation may also be regarded as **no bond resonance**



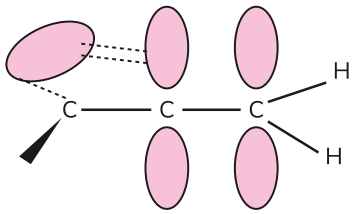


Figure 8.F Orbital diagram showing hyperconjugation in propene

Worked example 8.s

Why is $(CH_3)_3C^+$ more stable than $CH_3{}^+CH_2$, and ${}^+CH_3$ the least stable cation?

- the **hyperconjugation interaction in $(CH_3)_3C^+$ is greater than in $CH_3{}^+CH_2$**, the former holding **nine C-H bonds**
- in **${}^+CH_3$** the **vacant *p* orbital stands perpendicular to the plane the C-H bonds lie in**, so it **cannot overlap with them**; ${}^+CH_3$ therefore **lacks hyperconjugative stability**

Types of organic reaction and mechanism

- organic reactions fall into these categories
- **(i) Substitution** reactions
- **(ii) Addition** reactions
- **(iii) Elimination** reactions
- **(iv) Rearrangement** reactions
- these are studied in Unit 9 and later in class XII

C.1 Methods of purification of organic compounds

- whether an organic compound has been **taken from nature or made in the laboratory**, it has to be **cleaned up**
- the method chosen rests on the **nature of the compound and of the impurity present in it**
- the usual techniques: **(i) Sublimation · (ii) Crystallisation · (iii) Distillation · (iv) Differential extraction · (v) Chromatography**
- the last word on purity comes from the **melting or boiling point**: **a pure compound, as a rule, melts and boils sharply**
- newer methods of checking purity rest on different principles — continued on sheet 24

Sublimation

- heat some solids and they **go straight to vapour, skipping the liquid**
- the purification technique resting on that principle is **sublimation**, and it separates **sublimable compounds from non-sublimable impurities**

Crystallisation

- among the **commonest ways of cleaning up an organic solid**; it works because **compound and impurity dissolve to different extents in a well-chosen solvent**
- the solvent is one the compound **barely dissolves in cold but dissolves well in hot**
- the solution is **concentrated to a nearly saturated solution**; as it **cools the pure compound crystallises** and is **filtered off**
- the **filtrate, the mother liquor**, holds the impurities and a small quantity of the compound
- a compound **very soluble in one solvent and hardly at all in another** crystallises **well from a mixture of the pair**
- **activated charcoal adsorbs** any **coloured impurity**
- an impurity of **much the same solubility** as the compound calls for **crystallising more than once**

Distillation

- this important method separates **(i) volatile liquids from nonvolatile impurities** and **(ii) liquids whose boiling points differ sufficiently**
- liquids of unlike boiling point **turn to vapour at unlike temperatures**; each vapour is **cooled and its liquid collected on its own**
- this readily parts **chloroform (b.p 334 K)** from **aniline (b.p. 457 K)** (Figure 8.G)
- the mixture goes into a **round bottom flask** and is **heated with care**
- the **lower boiling component vaporises first**; they are **condensed in a condenser** and the liquid **collected in a receiver**
- the **vapours of the higher boiling component form later**, and that liquid is **collected separately**

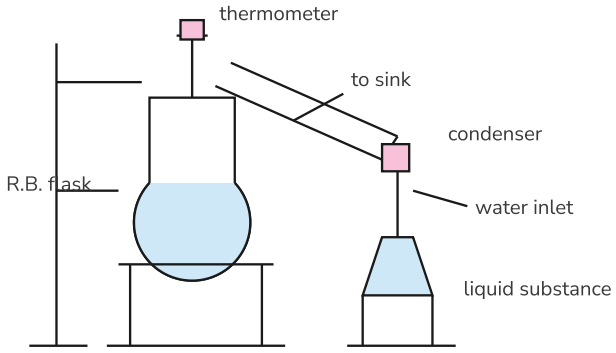


Figure 8.G Simple distillation. The vapours of a substance formed are condensed and the liquid is collected in conical flask.

Fractional distillation

- when **two liquids boil close together**, simple distillation **fails**: both vaporise **over the same range and condense together**
- **fractional distillation** is used in such cases: the **vapour goes up through a fractionating column before it is condensed**
- the column **sits in the neck of the flask** (Figure 8.H)
- the **vapours of the liquid with the higher boiling point condense before those of the lower boiling liquid**
- so the **vapours rising up the fractionating column become richer in the more volatile component** — continued on sheet 25

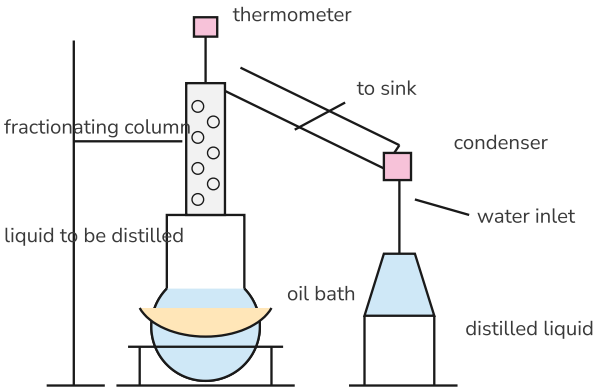


Figure 8.H Fractional distillation. The vapours of lower boiling fraction reach the top of the column first followed by vapours of higher boiling fractions.

- fractionating columns come in **various sizes and designs** (Figure 8.I)
- a column offers **many surfaces on which rising vapour and falling condensate can exchange heat**
- part of the condensate **draws heat from the vapour coming up and boils again**, so the **vapours grow richer in the low boiling component**
- the **vapours of the low boiling component ascend to the top**; there they are **pure in that component**, go through the condenser, and the **receiver collects the pure liquid**
- distillation after distillation, the **flask's contents grow richer in the high boiler**
- each **condense-and-revaporise stage in the column is one theoretical plate**; **industrial columns run to hundreds of plates**
- one technological application of fractional distillation is separating the different fractions of **crude oil in the petroleum industry**

Distillation under reduced pressure

- this method purifies liquids of **very high boiling point**, and those that **decompose at or below their boiling point**
- **lowering the pressure over the liquid makes it boil below its normal boiling point**
- boiling sets in once **vapour pressure matches the pressure outside**
- a **water pump or vacuum pump** does the reducing
- **glycerol** can be separated from **spent-lye in the soap industry** by this technique

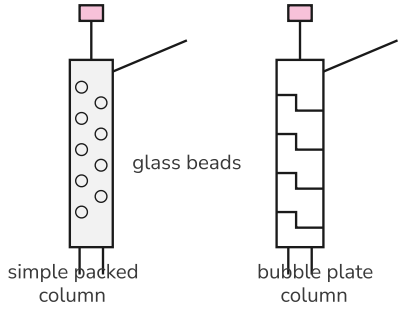


Figure 8.I Different types of fractionating columns.

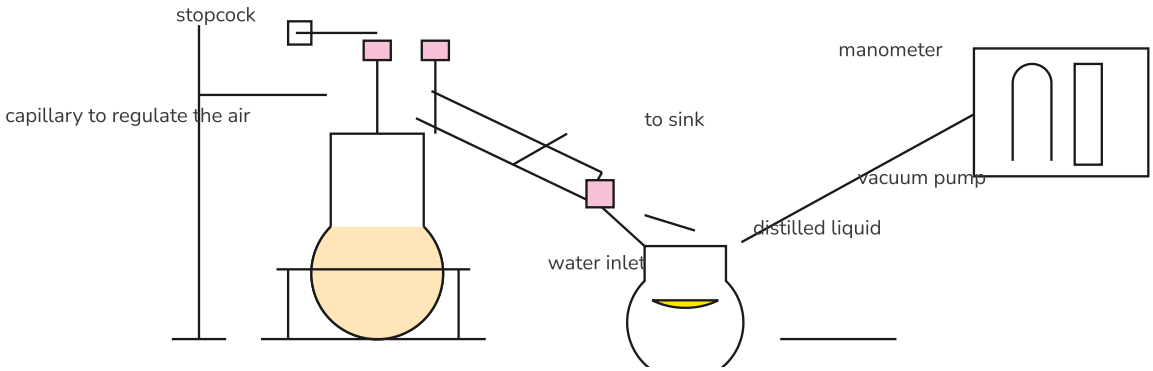


Figure 8.J Distillation under reduced pressure. A liquid boils at a temperature below its vapour pressure by reducing the pressure.

Steam distillation

- this technique separates substances that are **steam volatile and immiscible with water**
- **steam from a generator is blown through a heated flask of the liquid**
- the **steam carries the volatile compound over**; both are **condensed together**, and a **separating funnel parts the compound from the water afterwards**
- boiling starts when **the organic liquid's vapour pressure (p_1) and water's (p_2) together reach atmospheric pressure (p)**: $p = p_1 + p_2$
- **p_1 being lower than p, the organic liquid vaporises at a lower temperature than its boiling point**
- so a mixture of **water** with a **substance insoluble in it boils a little under 373K**
- what distils is water plus the substance, parted in a **separating funnel**; **aniline is separated this way from an aniline–water mixture** (Figure 8.K)

Differential extraction

- an organic compound **sitting in water** is drawn out by **shaking with an organic solvent it prefers to water**
- **solvent and water must not mix**, so that they **form two distinct layers a separatory funnel can separate**
- the organic solvent is then **removed by distillation or by evaporation to get the compound back**
- differential extraction is carried out in a **separatory funnel** (Figure 8.L)
- if the compound **dissolves poorly in the solvent**, **getting even a little of it out would take a great deal of solvent**
- then **continuous extraction** is the answer, the **one batch of solvent going round again and again**

Chromatography

- chromatography is an important technique, used extensively to **separate mixtures into their components**, to **purify compounds** and to **test the purity of compounds** — continued on sheet 27

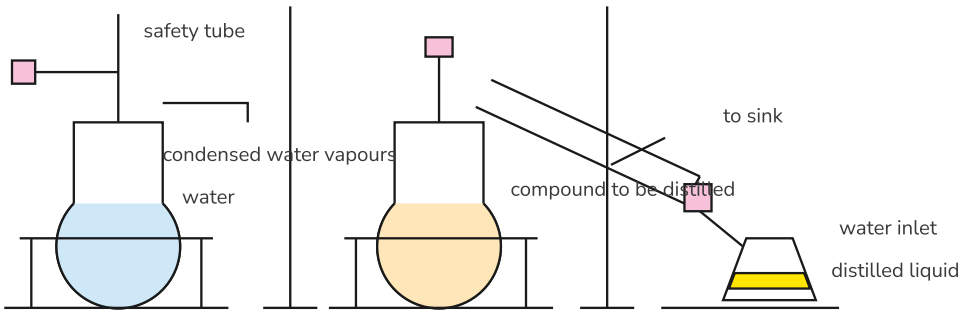


Figure 8.K Steam distillation. Steam volatile component volatilizes, the vapours condense in the condenser and the liquid collects in conical flask.

- the name **chromatography** rests on the Greek word ***chroma*, for colour**, the method having first been used to **separate coloured substances found in plants**
- the mixture is **loaded onto a stationary phase**, **solid or liquid**
- then a **solvent, a solvent mixture or a gas creeps across it**
- that moving stream is the **mobile phase**, and as it goes the **components drift apart**
- by the principle involved, chromatography falls into categories; two of them: **(a) adsorption** and **(b) partition chromatography**
- **(a) Adsorption chromatography** works because **an adsorbent holds unlike compounds unequally tightly**
- **silica gel and alumina** are the usual adsorbents
- as the **mobile phase passes over the adsorbent, each component is carried a different distance**
- the two main chromatographic techniques resting on **differential adsorption** are **(a) column chromatography** and **(b) thin layer chromatography**

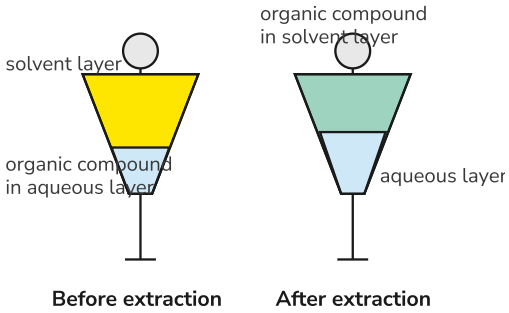


Figure 8.L Differential extraction. Extraction of compound takes place based on difference in solubility

Column chromatography

- here the stationary phase is **adsorbent packed into a glass tube**
- a **stopcock closes the bottom** (Figure 8.M)
- the **mixture adsorbed on adsorbent is placed on top of the adsorbent column**
- a suitable **eluant, one liquid or several, trickles down through it**
- **how strongly each compound is held decides where it ends up**, and the separation is complete
- the **most strongly adsorbed stay near the top**; the rest **travel further down, each a different way**

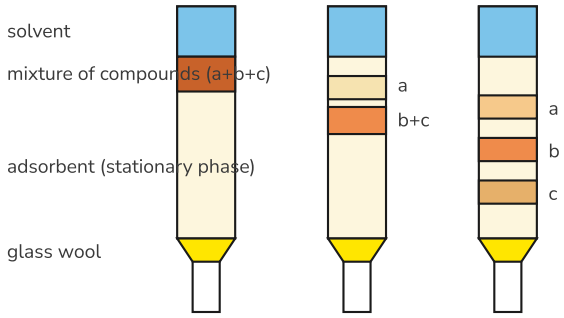


Figure 8.M Column chromatography. Different stages of separation of components of a mixture.

On a plate: TLC

- thin layer chromatography (TLC) is **adsorption chromatography** too, the adsorbent being a **thin coat on a glass plate**
- a **0.2mm layer** of **silica gel or alumina** is **spread on a glass plate of the right size**
- the plate is the **thin layer chromatography plate, or chromaplate**
- a **small spot of the mixture goes on about 2 cm from one end**
- the plate is then **placed in a closed jar holding the eluant** (Figure 8.N)
- the **eluant climbs the plate, carrying each component a different distance** according to **how strongly it is adsorbed**, and so they separate
- a component's **adsorption relative to the rest** is expressed as its **retardation factor, the R_f value**

$$R_f = \frac{\text{Distance moved by the substance from base line (x)}}{\text{Distance moved by the solvent from base line (y)}}$$

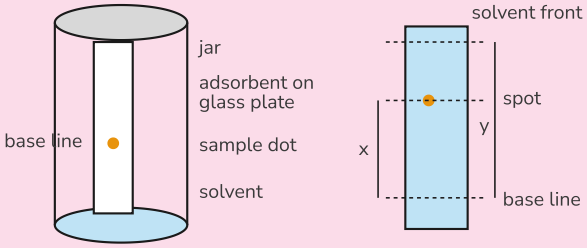


Figure 8.N (a) Thin layer chromatography. Chromatogram being developed.

(b) Developed chromatogram.

- the spots of **coloured compounds are visible on the TLC plate by their own colour**
- spots of **colourless compounds, invisible to the eye but fluorescing in ultraviolet light**, show up **under a UV lamp**
- or the plate goes into a **covered jar with a few iodine crystals**: compounds that **adsorb iodine show up as brown spots**
- sometimes an **appropriate reagent is sprayed on the plate: amino acids are detected by spraying with ninhydrin solution**

Partition chromatography

- it rests on the **continuous differential partitioning of the components of a mixture between a stationary and a mobile phase**
- **paper chromatography** is a type of partition chromatography, using a **special quality paper known as chromatography paper**
- that paper **holds water trapped in it, and the water acts as the stationary phase**
- a strip is **spotted near its foot with the mixture** and **hung with that end in a suitable solvent, or solvent mixture** (Figure 8.O), which serves as **mobile phase**
- **capillary action draws the solvent up through the spot**
- the paper **selectively retains the different components according to how they partition between the two phases**
- the strip so developed is a **chromatogram**
- **coloured compounds appear as spots at different heights above the start**
- **colourless ones** are found **under ultraviolet light**, or by an **appropriate spray reagent**, as under thin layer chromatography

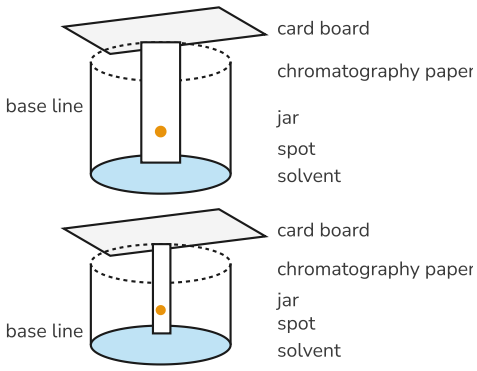


Figure 8.O Paper chromatography. Chromatography paper in two different shapes.

C.2 Qualitative analysis of organic compounds

- every organic compound holds **carbon and hydrogen**; **oxygen, nitrogen, sulphur, halogens and phosphorus** may join them
- **Detection of carbon and hydrogen** — **heat the compound with copper(II) oxide**
- **carbon becomes carbon dioxide**, which **turns lime-water turbid**
- **hydrogen becomes water**, which **turns anhydrous copper sulphate blue**

$$C + 2CuO \xrightarrow{\Delta} 2Cu + CO_2$$

$$2H + CuO \xrightarrow{\Delta} Cu + H_2O$$

$$CO_2 + Ca(OH)_2 \longrightarrow CaCO_3\downarrow + H_2O$$

$$5H_2O + CuSO_4 \longrightarrow CuSO_4.5H_2O$$

White *Blue*

Detection of other elements

- **"Lassaigne's test"** finds **nitrogen, sulphur, halogens and phosphorus**
- **fusing the compound with sodium metal turns its covalently bound elements into ions**

$$Na + C + N \xrightarrow{\Delta} NaCN$$

$$2Na + S \xrightarrow{\Delta} Na_2S$$

$$Na + X \xrightarrow{\Delta} NaX \quad (X = Cl, Br \text{ or } I)$$

- **C, N, S and X come from the organic compound**
- the **sodium cyanide, sulphide and halide** so made are **boiled out of the fused mass with distilled water**
- that extract is the **sodium fusion extract**

- **(A) Test for nitrogen** — **boil the extract with iron(II) sulphate, then acidify** — continued on sheet 30

- with **concentrated sulphuric acid**; a **Prussian blue colour proves nitrogen**
- first **iron(II) sulphate turns the cyanide into sodium hexacyanidoferrate(II)**
- hot concentrated sulphuric acid then **oxidises iron(II) to iron(III)**, and that combines with the hexacyanidoferrate(II) to give **iron(III) hexacyanidoferrate(II), ferriferrocyanide, the Prussian blue**

$$6CN^- + Fe^{2+} \longrightarrow [Fe(CN)_6]^{4-}$$

$$3[Fe(CN)_6]^{4-} + 4Fe^{3+} \xrightarrow{xH_2O} Fe_4[Fe(CN)_6]_3.xH_2O$$

Prussian blue

- where **both nitrogen and sulphur are present**, **sodium thiocyanate is formed**: it gives a **blood red colour instead of Prussian blue**, since **no cyanide is left free**

$$Na + C + N + S \longrightarrow NaSCN$$

$$Fe^{3+} + SCN^- \longrightarrow [Fe(SCN)]^{2+}$$

Blood red

- where the **sodium fusion is carried out with an excess of sodium**, the **thiocyanate decomposes to cyanide and sulphide**, and each then answers to its own test

$$NaSCN + 2Na \longrightarrow NaCN + Na_2S$$

(B) Test for sulphur

- **(a) acidify the extract with acetic acid and add lead acetate**; a **black lead sulphide precipitate means sulphur**

$$S^{2-} + Pb^{2+} \longrightarrow PbS$$

Black

- **(b)** treating the sodium fusion extract with **sodium nitroprusside**, the appearance of a **violet colour** further indicates sulphur

$$S^{2-} + [Fe(CN)_5NO]^{2-} \longrightarrow [Fe(CN)_5NOS]^{4-}$$

Violet

(C) Test for halogens

- **acidify the extract with nitric acid, then add silver nitrate**
- **white, dissolving in ammonium hydroxide: chlorine**
- **yellowish, barely dissolving: bromine**
- **yellow, not dissolving at all: iodine**

$$X^- + Ag^+ \longrightarrow AgX$$

X is a halogen — Cl, Br or I

- where **nitrogen or sulphur is also present**, the extract is **boiled with concentrated nitric acid beforehand** to **destroy the sodium cyanide or sulphide** the fusion made
- left in, those ions would **upset the silver nitrate test**

(D) Test for phosphorus

- **heating with sodium peroxide**, an oxidising agent, **turns the phosphorus to phosphate**
- after **boiling with nitric acid, ammonium molybdate goes in**
- **yellow, in solution or as a precipitate, means phosphorus**

$$Na_3PO_4 + 3HNO_3 \longrightarrow H_3PO_4 + 3NaNO_3$$

$$H_3PO_4 + 12(NH_4)_2MoO_4 + 21HNO_3 \longrightarrow (NH_4)_3PO_4.12MoO_3 + 21NH_4NO_3 + 12H_2O$$

ammonium molybdate → *ammonium phosphomolybdate*

C.3 Quantitative analysis

- quantitative analysis matters greatly in organic chemistry: it helps chemists **determine the mass per cent of the elements in a compound**
- Unit 1 showed that the **mass per cent of the elements is needed to determine the empirical and molecular formula**
- **Carbon and hydrogen** — both are **estimated in one experiment**
- a **known mass of the compound is burnt in the presence of excess oxygen and copper(II) oxide**, and the **carbon and hydrogen are oxidised to carbon dioxide and water**

$$C_xH_y + (x + y/4)\ O_2 \rightarrow x\ CO_2 + (y/2)\ H_2O$$

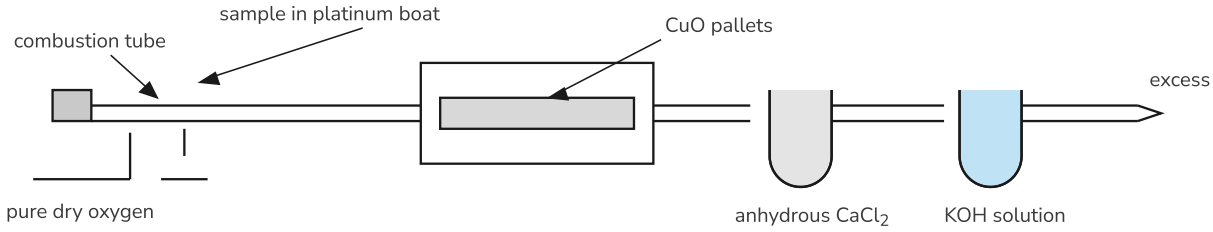


Figure 8.P Estimation of carbon and hydrogen. Water and carbon dioxide formed on oxidation of substance are absorbed in anhydrous calcium chloride and potassium hydroxide solutions respectively contained in U tubes.

- the **mass of water produced** is found by **passing the mixture through a weighed U-tube holding anhydrous calcium chloride**
- the **carbon dioxide is absorbed in another U-tube holding a concentrated solution of potassium hydroxide**, the two tubes being **connected in series** (Figure 8.P)
- **how much heavier each tube has grown tells how much water and carbon dioxide came off**, and the **carbon and hydrogen percentages follow**
- with **m g** of compound giving **m_1 g** of water and **m_2 g** of carbon dioxide:

$$\text{Percentage of carbon} = \frac{12 \times m_2 \times 100}{44 \times m}$$

$$\text{Percentage of hydrogen} = \frac{2 \times m_1 \times 100}{18 \times m}$$

Nitrogen

- nitrogen is estimated either by **(i) Dumas** or by **(ii) Kjeldahl**
- **(i) Dumas method — heat the compound with copper oxide under carbon dioxide**; out come **nitrogen gas, carbon dioxide and water**

$$C_xH_yN_z + (2x + y/2)\ CuO \longrightarrow x\ CO_2 + y/2\ H_2O + z/2\ N_2 + (2x + y/2)\ Cu$$

- **hot copper gauze reduces any trace of nitrogen oxide back to nitrogen**
- the gases are **collected over aqueous potassium hydroxide, which takes up the carbon dioxide**
- **nitrogen collects in the upper part of the graduated tube** (Figure 8.Q)

Worked example 8.t

Burning 0.246 g of an organic compound completely gave 0.198 g of carbon dioxide and 0.1014 g of water. Find the percentages of carbon and hydrogen.

$$\text{Percentage of carbon} = \frac{12 \times 0.198 \times 100}{44 \times 0.246} = 21.95\%$$

$$\text{Percentage of hydrogen} = \frac{2 \times 0.1014 \times 100}{18 \times 0.246} = 4.58\%$$

- let the mass of organic compound be **m g**, the volume of nitrogen collected **V_1 mL**, and the room temperature **T_1 K**

$$\text{Volume of nitrogen at STP} = \frac{P_1V_1 \times 273}{760 \times T_1} \text{ (let it be } V \text{ mL)}$$

- **p_1** and **V_1** belong to the **nitrogen collected**; **p_1 differs from the atmospheric pressure at which the nitrogen is collected**
- its value comes from **p_1 = Atmospheric pressure – Aqueous tension**
- **22400 mL N_2 at STP weighs 28 g**

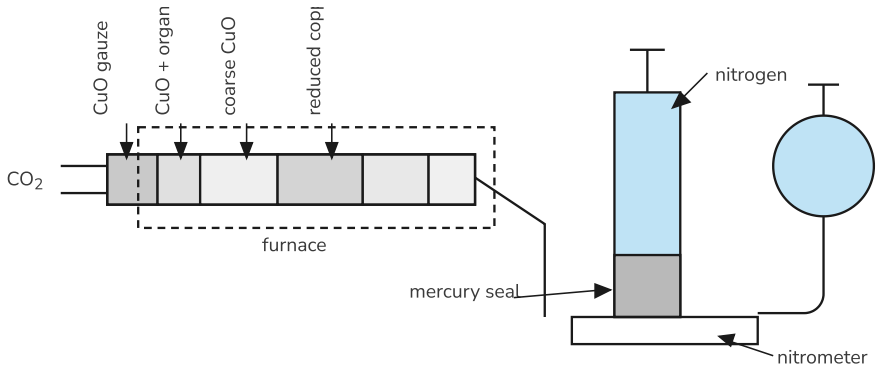


Figure 8.Q Dumas method. The organic compound yields nitrogen gas on heating it with copper(II) oxide in the presence of CO_2 gas. The mixture of gases is collected over potassium hydroxide solution in which CO_2 is absorbed and volume of nitrogen gas is determined.

$$V \text{ mL } N_2 \text{ at STP weighs} = \frac{28 \times V}{22400} \text{ g}$$

$$\text{Percentage of nitrogen} = \frac{28 \times V \times 100}{22400 \times m}$$

Worked example 8.u

By Dumas' method, 0.3 g of an organic compound yielded 50 mL of nitrogen, collected at 300 K and 715 mm. Aqueous tension at 300 K is 15 mm. What is the percentage of nitrogen?

- at 300 K the **50 mL** of nitrogen stands at a true pressure of **715 – 15 = 700 mm**

$$\text{Volume of nitrogen at STP} = \frac{273 \times 700 \times 50}{300 \times 760} = 41.9 \text{ mL}$$

- **22,400 mL of N_2 at STP weighs 28 g**

$$41.9 \text{ mL of nitrogen weighs} = \frac{28 \times 41.9}{22400} \text{ g}$$

$$\text{Percentage of nitrogen} = \frac{28 \times 41.9 \times 100}{22400 \times 0.3} = 17.46\%$$

Kjeldahl's method

- the compound holding nitrogen is **heated with concentrated sulphuric acid**, and the **nitrogen is converted to ammonium sulphate** (Figure 8.R)
- that acid mixture is then **warmed with excess sodium hydroxide**
- the **ammonia set free is caught in a known excess of standard sulphuric acid**
- the **acid the ammonia used up** measures **how much ammonia there was**
- to find it, **titrate the acid still unreacted against standard alkali**
- what counts is **how much of the acid first taken is gone** — continued on sheet 33

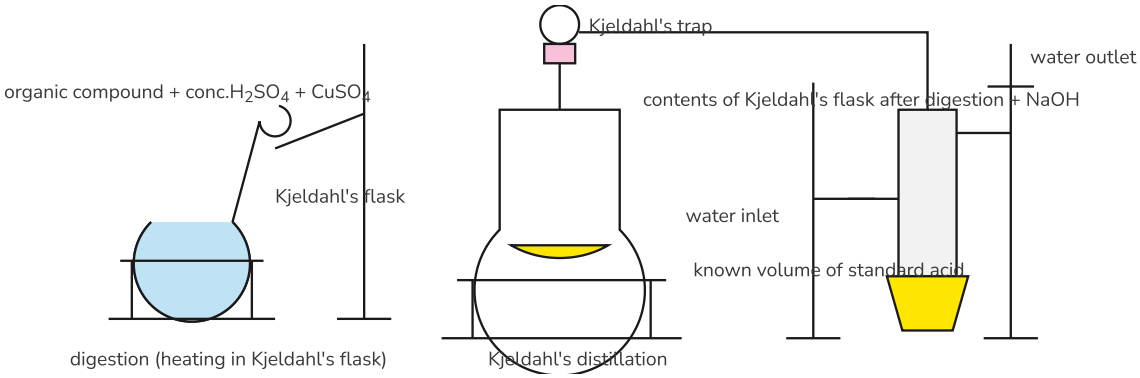


Figure 8.R Kjeldahl method. Nitrogen-containing compound is treated with concentrated H_2SO_4 to get ammonium sulphate which liberates ammonia on treating with NaOH; ammonia is absorbed in known volume of standard acid.

- **acid taken, less acid left over, is the acid the ammonia consumed**

Organic compound + H_2SO_4 ⟶ $(NH_4)_2SO_4$

$(NH_4)_2SO_4 \xrightarrow{2NaOH} Na_2SO_4 + 2NH_3 + 2H_2O$

$2NH_3 + H_2SO_4 \longrightarrow (NH_4)_2SO_4$

- let the mass of organic compound taken be **m g**
- volume of H_2SO_4 of molarity M taken = ***V* mL**
- volume of NaOH of molarity M used to titrate the excess H_2SO_4 = **V_1 mL**
- **V_1 mL of NaOH of molarity M = $V_1/2$ mL of H_2SO_4 of molarity M**
- volume of H_2SO_4 of molarity M unused = **$(V - V_1/2)$ mL**
- $(V - V_1/2)$ mL of H_2SO_4 of molarity M = **$2(V - V_1/2)$ mL of NH_3 solution of molarity M**
- **1000 mL of 1 M NH_3 solution contains 17 g NH_3, or 14 g of N**

$2(V - V_1/2)$ mL of NH_3 of molarity M contains $\frac{14 \times M \times 2(V - V_1/2)}{1000}$ g N

$$\text{Percentage of N} = \frac{14 \times M \times 2(V - V_1/2)}{1000} \times \frac{100}{m}$$

$$= \frac{1.4 \times M \times 2(V - V/2)}{m}$$

- Kjeldahl's method **does not apply to compounds holding nitrogen in nitro and azo groups**, nor to **nitrogen present in a ring**, pyridine for one
- nitrogen bound that way **never reaches ammonium sulphate under these conditions**

Worked example 8.v

By Kjeldahl's method, 0.5 g of a compound gave enough ammonia to neutralise 10 mL of 1 M H_2SO_4. What percentage of it is nitrogen?

- **1 M of 10 mL H_2SO_4 = 1 M of 20 mL NH_3**
- **1000 mL of 1M ammonia contains 14 g nitrogen**
- so **20 mL of 1M ammonia contains** the corresponding fraction — continued on sheet 34

Worked example 8.v · the answer

$$\frac{14 \times 20}{1000} \text{ g nitrogen}$$

$$\text{Percentage of nitrogen} = \frac{14 \times 20 \times 100}{1000 \times 0.5} = 56.0\%$$

Halogens · the Carius method

- a **weighed sample is heated with fuming nitric acid and silver nitrate** in a hard glass **Carius tube** (Figure 8.S)
- the tube is heated **in a furnace**
- **carbon and hydrogen go to carbon dioxide and water**
- the **halogen ends up as silver halide (AgX)**, which is **filtered off, washed, dried and weighed**
- with **m g** of compound giving **m_1 g** of AgX:
- **1 mol of AgX contains 1 mol of X**

$$\text{Mass of halogen in } m_1 \text{ g of AgX} = \frac{\text{atomic mass of X} \times m_1 \text{ g}}{\text{molecular mass of AgX}}$$

$$\text{Percentage of halogen} = \frac{\text{atomic mass of X} \times m_1 \text{ g}}{\text{molecular mass of AgX}}$$

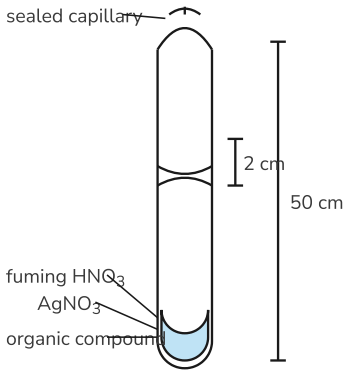


Figure 8.S Carius method. Halogen containing organic compound is heated with fuming nitric acid in the presence of silver nitrate.

Worked example 8.w

A Carius estimation on 0.15 g of an organic compound yielded 0.12 g of AgBr. What percentage of bromine does the compound hold?

- molar mass of AgBr = **108 + 80 = 188 g mol^{-1}**
- **80 g of bromine sits in every 188 g AgBr**

$$0.12 \text{ g AgBr contains } \frac{80 \times 0.12}{188} \text{ g bromine}$$

$$\text{Percentage of bromine} = \frac{80 \times 0.12 \times 100}{188 \times 0.15} = 34.04\%$$

Sulphur

- a **weighed sample goes into a Carius tube with sodium peroxide or fuming nitric acid and is heated**
- the **sulphur goes to sulphuric acid**
- **excess aqueous barium chloride brings it down as barium sulphate**
- the precipitate is **filtered off, washed, dried and weighed**, and its **mass gives the sulphur percentage**
- with **m g** of compound giving **m_1 g** of barium sulphate:
- **1 mol of $BaSO_4$ = 233 g $BaSO_4$ = 32 g sulphur**

$$m_1 \text{ g } BaSO_4 \text{ contains } \frac{32 \times m_1}{233} \text{ g sulphur}$$

$$\text{Percentage of sulphur} = \frac{32 \times m_1 \times 100}{233 \times m}$$

Worked example 8.x

0.157 g of an organic compound yielded 0.4813 g of barium sulphate. Find the sulphur percentage.

- molecular mass of $BaSO_4$ = **137 + 32 + 64 = 233 g**
- **32 g of sulphur sits in every 233 g $BaSO_4$**

$$0.4813 \text{ g } BaSO_4 \text{ contains } \frac{32 \times 0.4813}{233} \text{ g sulphur}$$

$$\text{Percentage of sulphur} = \frac{32 \times 0.4813 \times 100}{233 \times 0.157} = 42.10\%$$

Phosphorus

- a **weighed sample is heated with fuming nitric acid**, which **takes the phosphorus to phosphoric acid**
- **ammonia and ammonium molybdate bring it down as ammonium phosphomolybdate, $(NH_4)_3PO_4.12MoO_3$**
- or **magnesia mixture brings it down as $MgNH_4PO_4$**, which **ignites to $Mg_2P_2O_7$**
- with **m g** of compound giving **m_1 g** of ammonium phospho molydate:
- molar mass of $(NH_4)_3PO_4.12MoO_3$ = **1877 g**

$$\text{Percentage of phosphorus} = \frac{31 \times m_1 \times 100}{1877 \times m} \%$$

- where phosphorus is estimated as $Mg_2P_2O_7$:

$$\text{Percentage of phosphorus} = \frac{62 \times m_1 \times 100}{222}$$

- here **222 u is $Mg_2P_2O_7$'s molar mass**, **m the sample mass**, **m_1 the $Mg_2P_2O_7$ obtained**, and **62 the mass of its two phosphorus atoms**, in $Mg_2P_2O_7$

Oxygen

- oxygen is usually found **by subtraction: 100 less the percentages of every other element**
- oxygen can nevertheless be **estimated directly**, as follows
- a weighed sample is **broken down by heat in a nitrogen stream**
- the **oxygen-bearing gases pass over red-hot coke**, which **turns every bit of oxygen into carbon monoxide**
- the mixture is then **passed through warm iodine pentoxide (I_2O_5)**, which **oxidises the carbon monoxide to carbon dioxide and sets iodine free**

$$\text{Compound} \xrightarrow{\text{heat}} O_2 + \text{other gaseous products}$$

$$2C + O_2 \xrightarrow{1373\text{ K}} 2CO] \times 5 \quad \text{(A)}$$

$$I_2O_5 + 5CO \longrightarrow I_2 + 5CO_2] \times 2 \quad \text{(B)}$$

- making the **amount of CO produced in (A) equal to the amount used in (B)**, by multiplying (A) and (B) by **5 and 2**, it appears that **every mole of oxygen the compound gives up ends as two moles of carbondioxide**
- so **32 g of oxygen shows up as 88 g of carbon dioxide**
- with ***m* g** of compound giving **m_1 g** of carbon dioxide:

$$m_1 \text{ g carbon dioxide is obtained from } \frac{32 \times m_1}{88} \text{ g } O_2$$

$$\text{Percentage of oxygen} = \frac{32 \times m_1 \times 100}{88 \times m} \%$$

- the **iodine set free** would serve just as well for the oxygen percentage
- today elements are estimated on **microquantities, by automated methods**
- the **carbon, hydrogen and nitrogen** present are determined by an apparatus known as a **CHN elemental analyser**
- it takes **a mere 1-3 mg** and **puts the figures on a screen in a short time**

Summary

- this unit has covered the basics of **how organic compounds are built and how they react**; **covalent bonding** holds them together
- the nature of that bonding is described through the **concept of orbital hybridisation**, by which carbon may hold **sp^3, sp^2 and sp hybridised orbitals**
- those three are found in **methane, ethene and ethyne** respectively, and the concept explains the **tetrahedral shape of methane, the planar shape of ethene and the linear shape of ethyne**
- an **sp^3 hybrid orbital overlapping with the 1s orbital of hydrogen gives a carbon–hydrogen (C–H) single bond, a sigma (σ) bond**
- overlap of an **sp^2 orbital of one carbon with that of another gives a carbon–carbon σ bond**
- a **pi (π) bond comes of two neighbouring carbons' unhybridised p orbitals overlapping sideways**
- organic compounds may be written by **various structural formulas**, and their **three dimensional structure is put on paper by the wedge and dash formula**

- organic compounds are classified by their **structure** or by the **functional groups they hold**
- a **functional group, an atom or cluster of atoms bonded in a particular way, fixes a compound's physical and chemical behaviour**
- names follow the **IUPAC** rules, under which the **name is correlated with the structure so that the reader can deduce one from the other**

- mechanism rests on five things: the **substrate's structure**, **how the covalent bond breaks**, the **attacking reagent**, the **electron displacement effects**, and the **reaction conditions**
- every reaction **breaks covalent bonds and makes new ones**; a bond can break **heterolytically or homolytically**
- **heterolytic cleavage yields carbocations or carbanions**; **homolytic cleavage gives free radicals** as the reactive intermediate
- heterolytic reactions pair off **two complementary species**: the **nucleophile, which gives an electron pair**, and the **electrophile, which takes it**
- the **inductive, resonance, electromeric and hyperconjugation effects** can **polarise a bond**, so that **particular carbons or other atoms become electron-poor or electron-rich**
- the four broad reaction types are **substitution, addition, elimination and rearrangement**

- to **settle a structure** one **purifies, then analyses qualitatively and quantitatively**
- the methods of purification — **sublimation, distillation and differential extraction** — all turn on **some physical property in which the components differ**
- **chromatography** serves to **separate, identify and purify**; it comes in two kinds, **adsorption** and **partition**
- **adsorption chromatography** uses the **unequal grip an adsorbent has on unlike components**
- **partition chromatography** involves the **continuous partitioning of the components between stationary and mobile phases**
- once the compound is **pure**, its **qualitative analysis detects the elements present in it**
- **Lassaigne's test finds nitrogen, sulphur, halogens and phosphorus**
- **carbon and hydrogen are weighed as the carbon dioxide and water they burn to**
- **nitrogen goes by Dumas or Kjeldahl, halogens by Carius**
- **sulphur and phosphorus are first oxidised, to sulphuric and phosphoric acid**
- **oxygen is usually what is left when every other element's percentage is taken from 100**

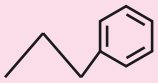
Exercises

E1 Give the hybridisation state of each carbon atom in $CH_2{=}C{=}O$, $CH_3CH{=}CH_2$, $(CH_3)_2CO$, $CH_2{=}CHCN$ and C_6H_6.

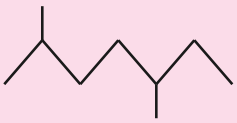
E2 Mark the σ and the π bonds in C_6H_6, C_6H_{12}, CH_2Cl_2, $CH_2{=}C{=}CH_2$, CH_3NO_2 and $HCONHCH_3$.

E3 Draw isopropyl alcohol, 2,3-dimethylbutanal and heptan-4-one as bond line formulas.

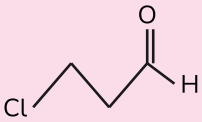
E4 Give the IUPAC name of each compound below.

(a) 

(b) 

(c) 

(d) 

(e) 

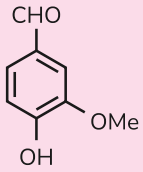
(f) Cl_2CHCH_2OH

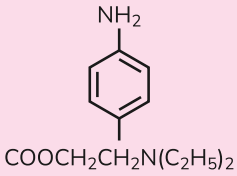
E5 Which of each pair is the correct IUPAC name for the compound concerned? (a) 2,2-Dimethylpentane or 2-Dimethylpentane; (b) 2,4,7-Trimethyloctane or 2,5,7-Trimethyloctane; (c) 2-Chloro-4-methylpentane or 4-Chloro-2-methylpentane; (d) But-3-yn-1-ol or But-4-ol-1-yne

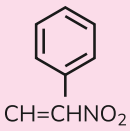
E6 Starting from each of these, draw the first five members of its homologous series: (a) H–COOH, (b) CH_3COCH_3, (c) H–CH=CH_2.

E7 For each of the following give a condensed formula and a bond line formula, and name any functional group present: (a) 2,2,4-Trimethylpentane; (b) 2-Hydroxy-1,2,3-propanetricarboxylic acid; (c) Hexanedial.

E8 Identify the functional groups in the compounds below.

(a) 

(b) 

(c) 

E9 Which of $O_2NCH_2CH_2O^-$ and $CH_3CH_2O^-$ would you expect to be the more stable, and why?

E10 Attached to a π system, an alkyl group donates electrons. Why?

E11 Draw the resonance structures of each compound below, showing the electron shift by curved-arrow notation. (a) C_6H_5OH; (b) $C_6H_5NO_2$; (c) $CH_3CH{=}CHCHO$; (d) C_6H_5–CHO; (e) $C_6H_5{-}{}^+CH_2$; (f) $CH_3CH{=}CH{-}{}^+CH_2$

E12 Define electrophile and nucleophile, giving examples of each.

E13 Identify each reagent shown in bold as a nucleophile or an electrophile: (a) $CH_3COOH + \mathbf{HO^-} \rightarrow CH_3COO^- + H_2O$

E13 (continued) (b) $CH_3COCH_3 + \mathbf{C^-N} \rightarrow (CH_3)_2C(CN)(OH)$; (c) $C_6H_6 + \mathbf{CH_3C^+O} \rightarrow C_6H_5COCH_3$

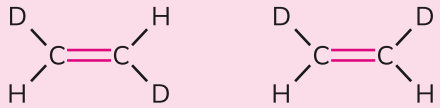
E14 Put each reaction into one of the reaction types taken in this unit. (a) $CH_3CH_2Br + HS^- \rightarrow CH_3CH_2SH + Br^-$; (b) $(CH_3)_2C{=}CH_2 + HCl \rightarrow (CH_3)_2ClC{-}CH_3$; (c) $CH_3CH_2Br + HO^- \rightarrow CH_2{=}CH_2 + H_2O + Br^-$; (d) $(CH_3)_3C{-}CH_2OH + HBr \rightarrow (CH_3)_2CBrCH_2CH_2CH_3 + H_2O$

E15 How do the members of each pair of structures stand to one another? Are they structural isomers, geometrical isomers, or resonance contributors?

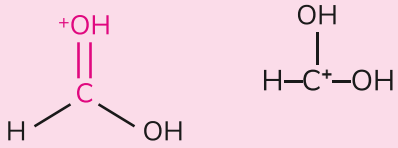
(a)



(b)



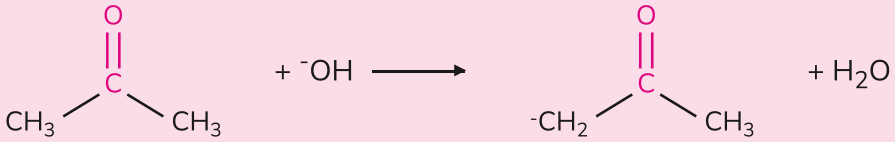
(c)



E16 For each bond cleavage below, trace the electron flow with curved arrows and say whether it is homolysis or heterolysis. Identify the reactive intermediate produced as free radical, carbocation or carbanion.

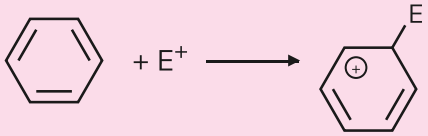
(a) $CH_3O{-}OCH_3 \longrightarrow CH_3\dot{O} + \dot{O}CH_3$

(b)



(c) $(CH_3)_3C{-}Br \longrightarrow (CH_3)_3C^+ + Br^-$

(d)



E17 Explain the terms inductive and electromeric effect. Each order of carboxylic acid acidity below is correct. Which electron displacement effect accounts for it? (a) $Cl_3CCOOH > Cl_2CHCOOH > ClCH_2COOH$; (b) $CH_3CH_2COOH > (CH_3)_2CHCOOH > (CH_3)_3C.COOH$

E18 Give a brief description of the principle behind each technique, with an example in each case: (a) Crystallisation; (b) Distillation; (c) Chromatography

E19 Describe a method that will separate two compounds of different solubility in a solvent S.

E20 Distinguish simple distillation, reduced-pressure distillation and steam distillation.

E21 Discuss the chemistry of Lassaigne's test.

E22 Set out the difference in principle between estimating nitrogen by (i) the Dumas method and (ii) Kjeldahl's method.

E23 Discuss the principle behind estimating the halogens, sulphur and phosphorus present in an organic compound.

E24 Explain the principle of paper chromatography.

E25 Why is nitric acid added to the sodium extract before silver nitrate, when testing for halogens?

E26 Why is an organic compound fused with metallic sodium before testing for nitrogen, sulphur and halogens?

E27 Name a suitable technique for separating the components of a mixture of calcium sulphate and camphor.

E28 Why does an organic liquid vaporise at a temperature below its boiling point in steam distillation?

E29 Will CCl_4 give a white precipitate of AgCl on heating with silver nitrate? Give your reason.

E30 Why is a solution of potassium hydroxide used to absorb the carbon dioxide evolved while estimating the carbon in an organic compound?

E31 Why must acetic acid, and not sulphuric acid, be used to acidify the sodium extract when testing for sulphur by the lead acetate test?

E32 An organic compound is 69% carbon and 4.8% hydrogen, the rest oxygen. How much carbon dioxide and how much water will 0.20 g of it give on complete combustion?

E33 A 0.50 g sample of an organic compound was treated by Kjeldahl's method. The ammonia given off was taken up in 50 mL of 0.5 M H_2SO_4; neutralising the acid left over took 60 mL of 0.5 M NaOH. What is the compound's nitrogen percentage?

E34 In a Carius estimation, 0.3780 g of an organic chloro compound gave 0.5740 g of silver chloride. Calculate the percentage of chlorine in the compound.

E35 A Carius estimation on 0.468 g of an organic sulphur compound gave 0.668 g of barium sulphate. What percentage of the compound is sulphur?

E36 In the organic compound $CH_2{=}CH{-}CH_2{-}CH_2{-}C{\equiv}CH$, which pair of hybridised orbitals forms the C2–C3 bond? (a) sp–sp^2; (b) sp–sp^3; (c) sp^2–sp^3; (d) sp^3–sp^3

E37 In Lassaigne's test for nitrogen, the Prussian blue colour is obtained from the formation of which compound? (a) $Na_4[Fe(CN)_6]$; (b) $Fe_4[Fe(CN)_6]_3$; (c) $Fe_2[Fe(CN)_6]$; (d) $Fe_3[Fe(CN)_6]_4$

E38 Which of these carbocations is the most stable? (a) $(CH_3)_3C.^+CH_2$; (b) $(CH_3)_3C^+$; (c) $CH_3CH_2{}^+CH_2$; (d) $CH_3{}^+CH\ CH_2CH_3$

E39 The best and latest technique for isolating, purifying and separating organic compounds is: (a) Crystallisation; (b) Distillation; (c) Sublimation; (d) Chromatography

E40 The reaction $CH_3CH_2I + KOH(aq) \rightarrow CH_3CH_2OH + KI$ is classed as: (a) electrophilic substitution; (b) nucleophilic substitution; (c) elimination; (d) addition